Name: Mahita Suresh Thundiyil | **PRN:** 24070126102 | **Batch/Branch/Div:** 2024-28 B.Tech.AIML B2

Data Visualization Lab

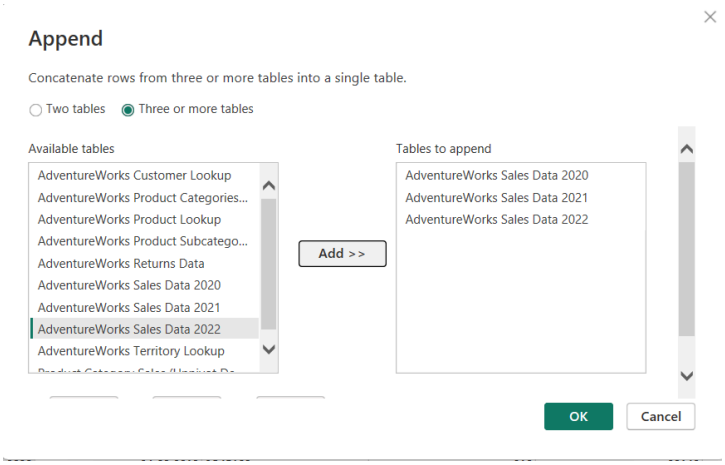
Assignment 4: Use the Adventure Works dataset and select the appropriate worksheet(s) to complete the given tasks.

Application Used: Microsoft Power BI

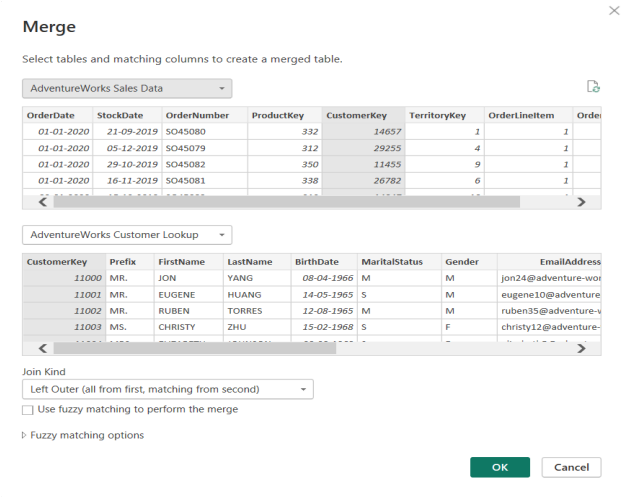
Before we start we need to append the “AdventureWorks Sales Data 2020”, “AdventureWorks Sales Data 2021”, and “AdventureWorks Sales Data 2022” to create a single table called “AdventureWorks Sales Data”.

For this I did the following:

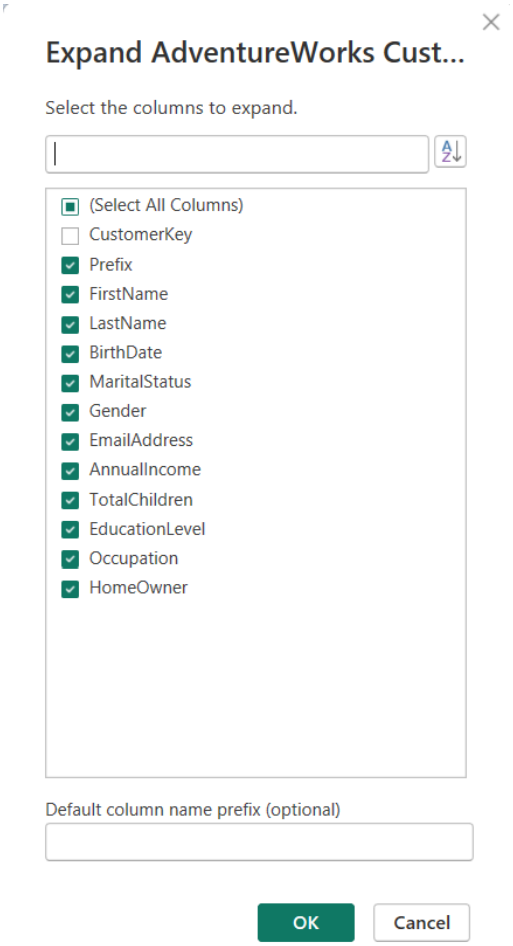
- Open Power Query in “Transform” mode by clicking “Transform” from the “Home” tab.
- Clicked on “Append” with the “Append as New Query” option.
- Selected all 3 datasets.
- Clicked Ok.
- Renamed the new table as “AdventureWorks Sales Data”.



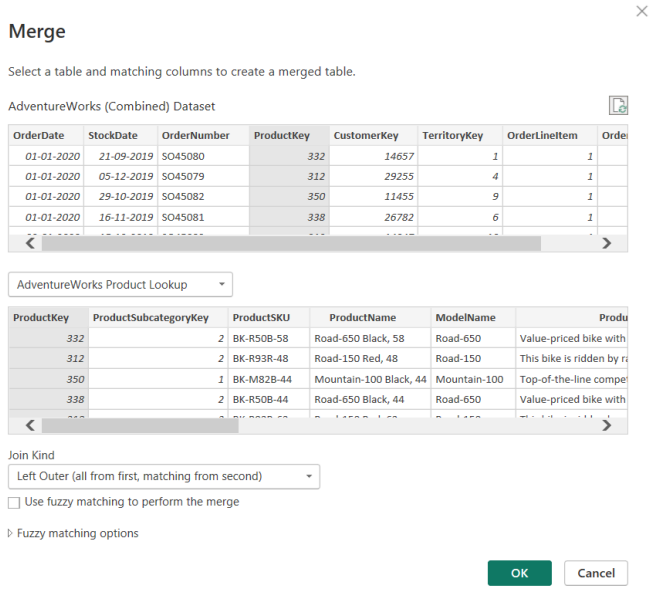
- I’ve created a single dataset for most of my queries - “AdventureWorks (Combined) Dataset”.
- Merged "AdventureWorks Sales Data” table with "AdventureWorks Customer Lookup” table as a new dataset “AdventureWorks (Combined) Dataset” through “CustomerKey”.



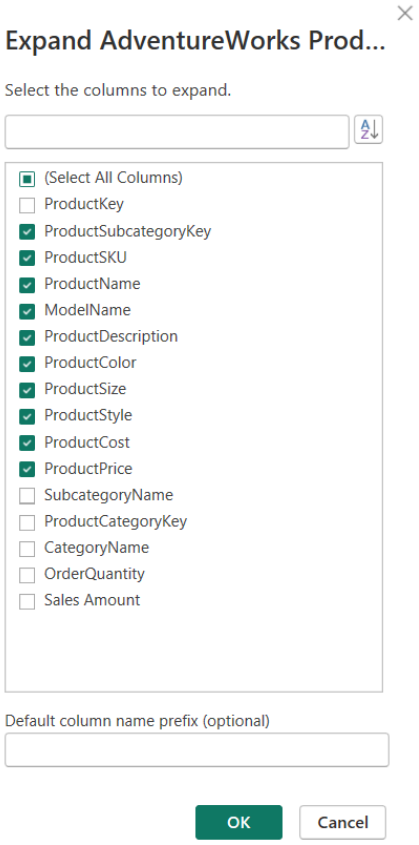
- Expanded columns - Prefix, FirstName, LastName, BirthDate, MartialStatus, Gender, EmailAddress, AnnualIncome, TotalChildren, EducationalLevel, Occupation, and HomeOwner.



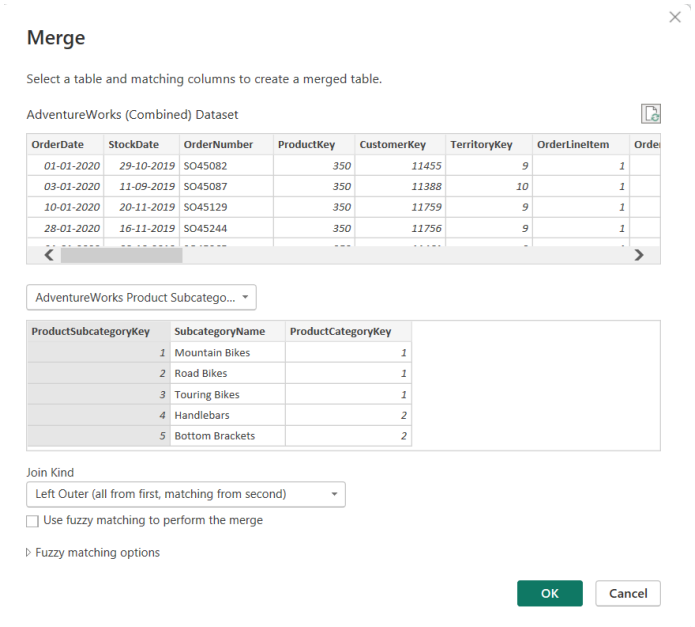
- Merged "AdventureWorks (Combined) Dataset" table with "AdventureWorks Product Lookup" table through "ProductKey".



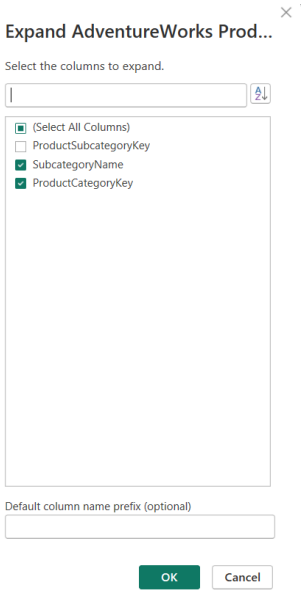
- Expanded - ProductSucategoryKey, ProductSKU, ProductName, ModelName, ProductDescription, ProductColor, ProductSize, ProductStyle, ProductCost, ProductPrice.



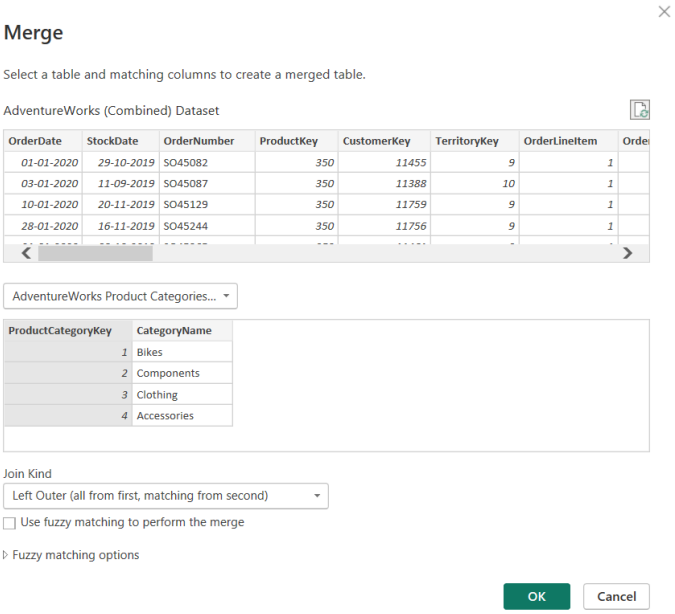
- Merged "AdventureWorks (Combined) Dataset" table with "AdventureWorks Product Subcategories Lookup" table through "ProductSubcategoryKey".



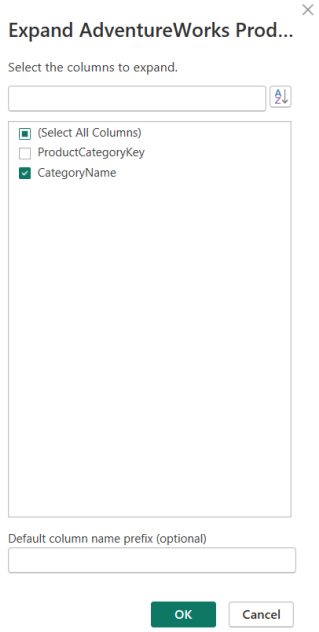
- Expanded columns - SubcategoryName, and ProductCategoryKey



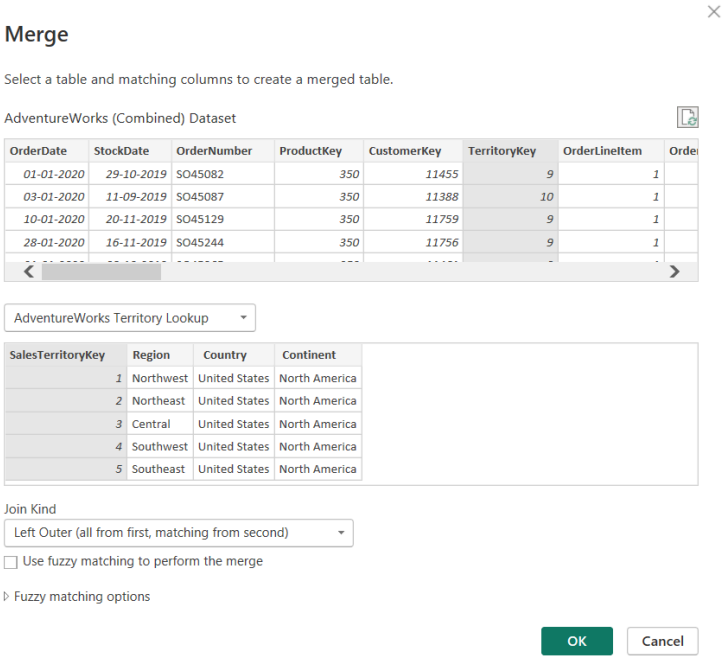
- Merged "AdventureWorks (Combined) Dataset" table with "AdventureWorks Product Categories Lookup" table through "ProductCategoryKey".



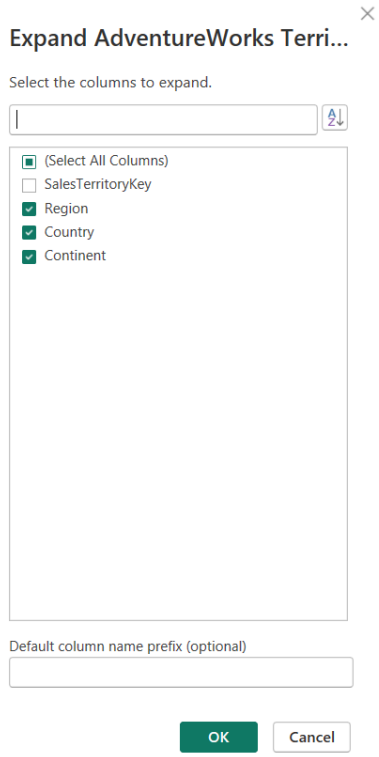
- Expanded column - CategoryName.



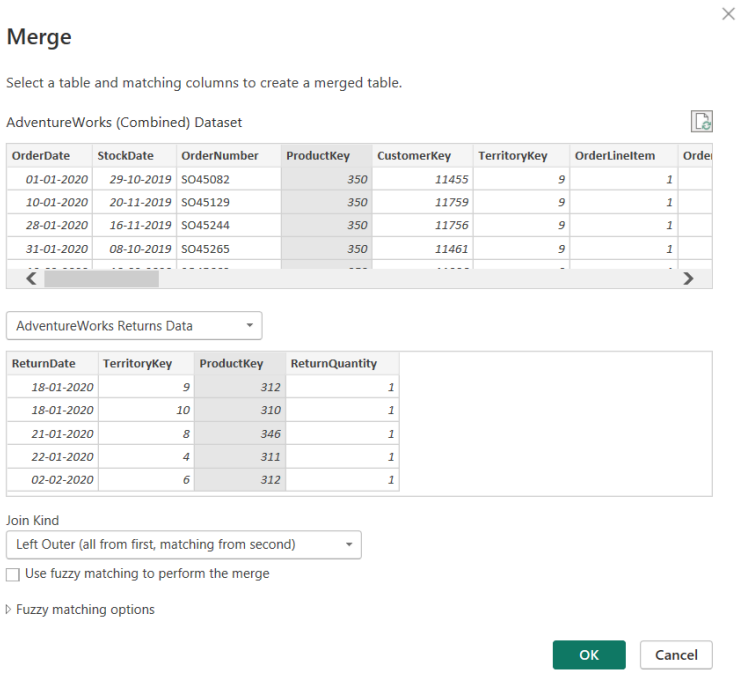
- Merged "AdventureWorks (Combined) Dataset" table with "AdventureWorks Territory Lookup" table through "TerritoryKey".



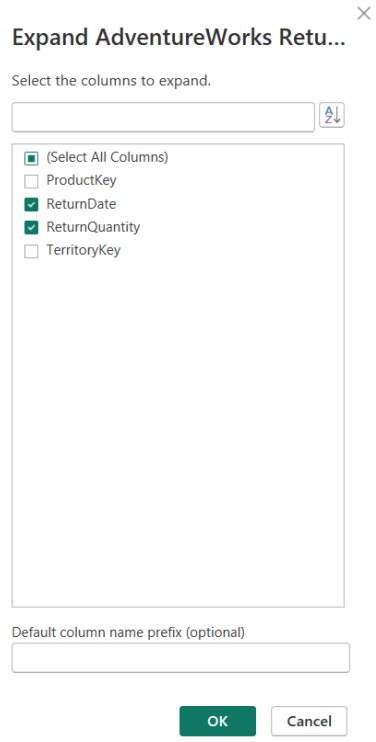
- Expanded columns - Region, Country, Continent.



- Merged "AdventureWorks (Combined) Dataset" table with "AdventureWorks Returns Data Lookup" table through "ProductKey".



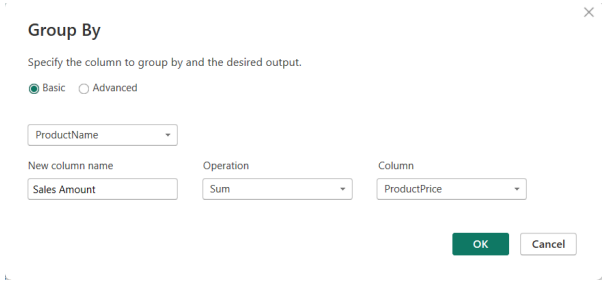
- Expanded columns -



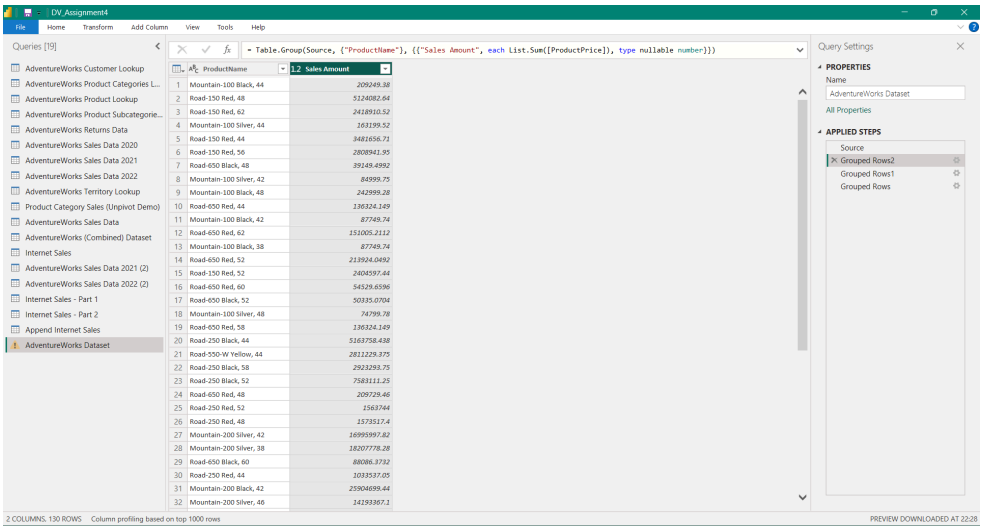
Part A: Power Query Operations

A. Grouping

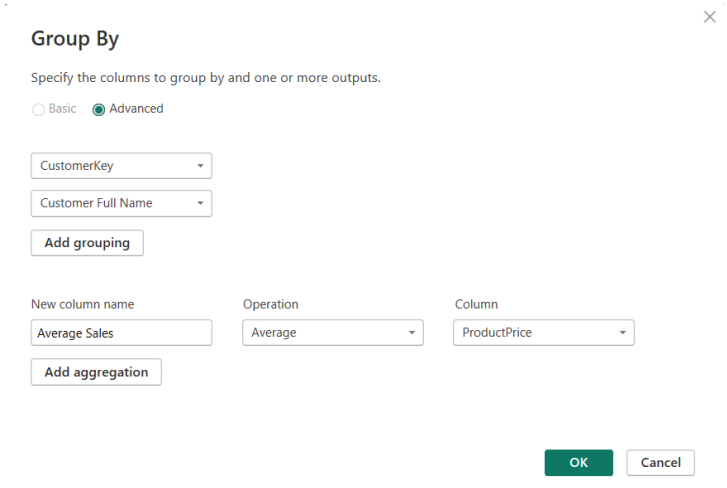
1. Find the total sales amount for each product category.
    - Click on GroupBy which is on the Home / Transform tab. Select ‘ProductName’ for grouping ‘Sales Amount’.
    - Sales Amount = SUM([ProductPrice])



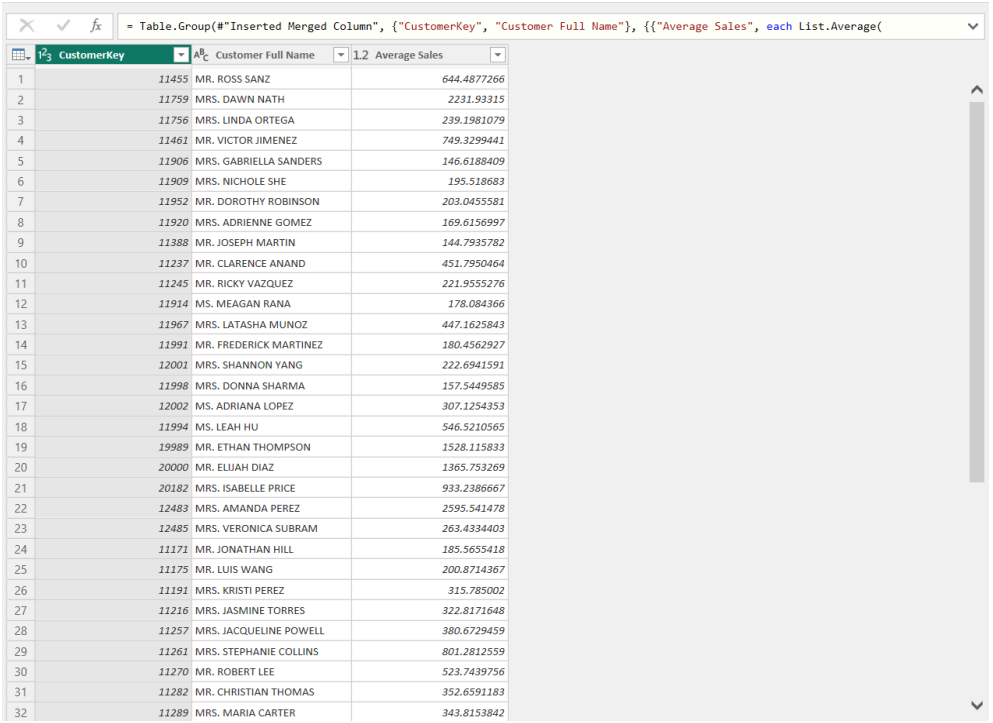
- Here is the result:



2. Calculate the average sales amount for each customer.
    - Click on GroupBy which is on the Home / Transform tab. Go to ‘Advanced’ mode. Select ‘CustomerKey’ and ‘Customer Full Name’ for grouping ‘Average Sales’.
    - Average Sales = AVERAGE([ProductPrice])



- Here is the result:

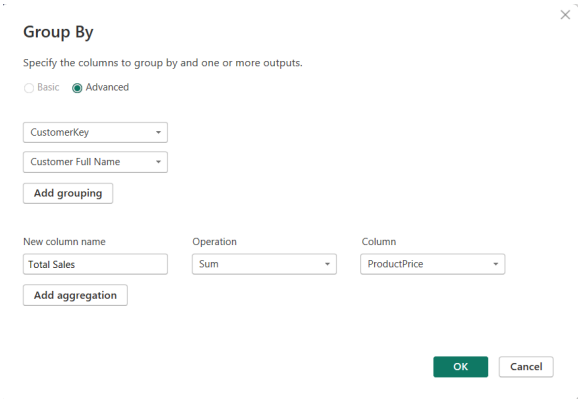


= Table.Group(#"Inserted Merged Column", {"CustomerKey", "Customer Full Name"}, {{"Average Sales", each List.Average(

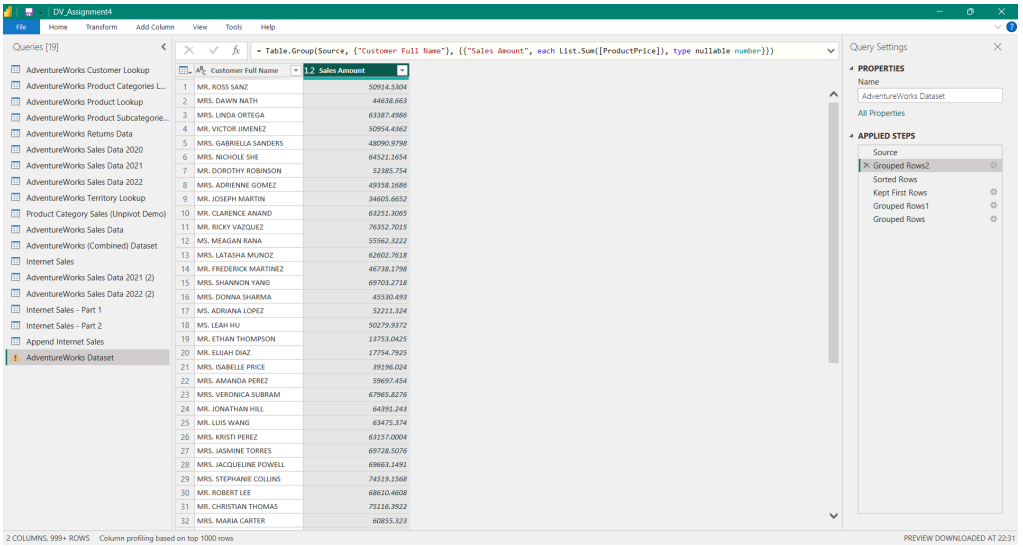
	CustomerKey	Customer Full Name	Average Sales
1	11455	MR. ROSS SANZ	644.4877266
2	11759	MRS. DAWN NATH	2231.93315
3	11756	MRS. LINDA ORTEGA	239.1981079
4	11461	MR. VICTOR JIMENEZ	749.3299441
5	11906	MRS. GABRIELLA SANDERS	146.6188409
6	11909	MRS. NICHOLE SHE	195.518683
7	11952	MR. DOROTHY ROBINSON	203.0455581
8	11920	MRS. ADRIENNE GOMEZ	169.6156997
9	11388	MR. JOSEPH MARTIN	144.7935782
10	11237	MR. CLARENCE ANAND	451.7950464
11	11245	MR. RICKY VAZQUEZ	221.9555276
12	11914	MS. MEAGAN RANA	178.084366
13	11967	MRS. LATASHA MUNOZ	447.1625843
14	11991	MR. FREDERICK MARTINEZ	180.4562927
15	12001	MRS. SHANNON YANG	222.6941591
16	11998	MRS. DONNA SHARMA	157.5449585
17	12002	MS. ADRIANA LOPEZ	307.1254353
18	11994	MS. LEAH HU	546.5210565
19	19989	MR. ETHAN THOMPSON	1528.115833
20	20000	MR. ELIJAH DIAZ	1365.753269
21	20182	MRS. ISABELLE PRICE	933.2386667
22	12483	MRS. AMANDA PEREZ	2595.541478
23	12485	MRS. VERONICA SUBRAM	263.4334403
24	11171	MR. JONATHAN HILL	185.5655418
25	11175	MR. LUIS WANG	200.8714367
26	11191	MRS. KRISTI PEREZ	315.785002
27	11216	MRS. JASMINE TORRES	322.8171648
28	11257	MRS. JACQUELINE POWELL	380.6729459
29	11261	MRS. STEPHANIE COLLINS	801.2812559
30	11270	MR. ROBERT LEE	523.7439756
31	11282	MR. CHRISTIAN THOMAS	352.6591183
32	11289	MRS. MARIA CARTER	343.8153842

3. Identify the top 10 customers based on total sales.

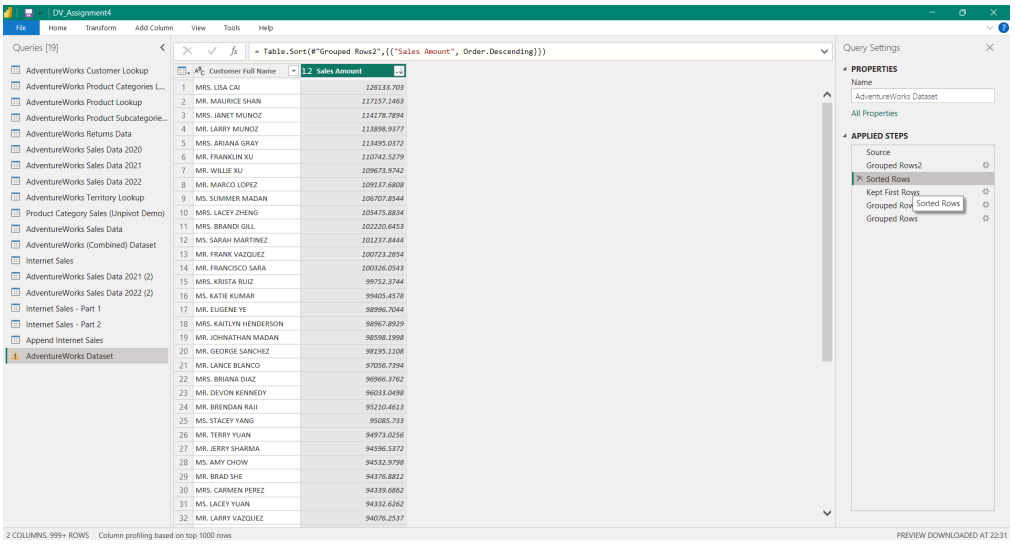
- Click on GroupBy which is on the Home / Transform tab. Go to ‘Advanced’ mode. Select ‘CustomerKey’ and ‘Customer Full Name’ for grouping ‘Total Sales’.
- Total Sales = SUM([ProductPrice])



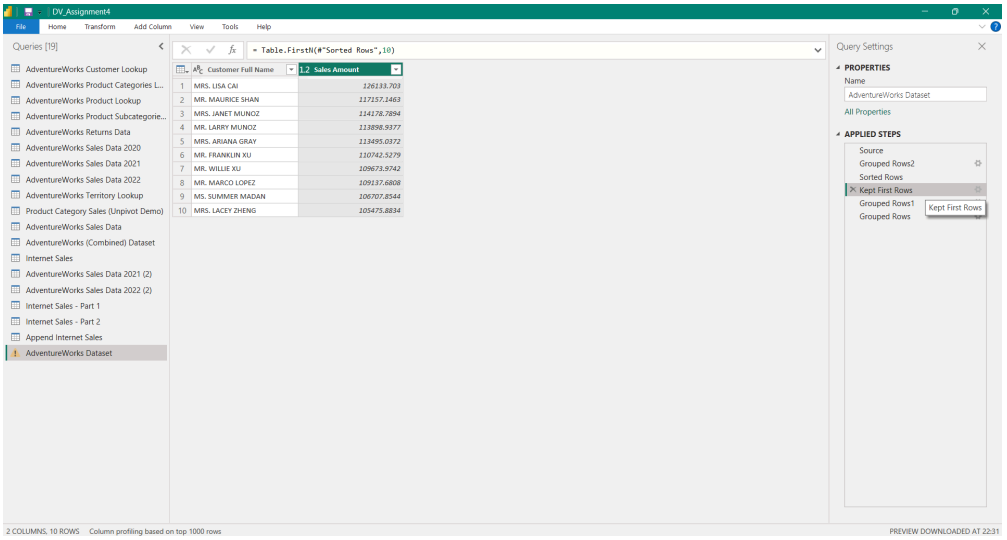
- Here is the result:



- Now, sort it in descending order by clicking on ‘Sales Amount’ and clicking on Sort (Z-A) from the Home tab.

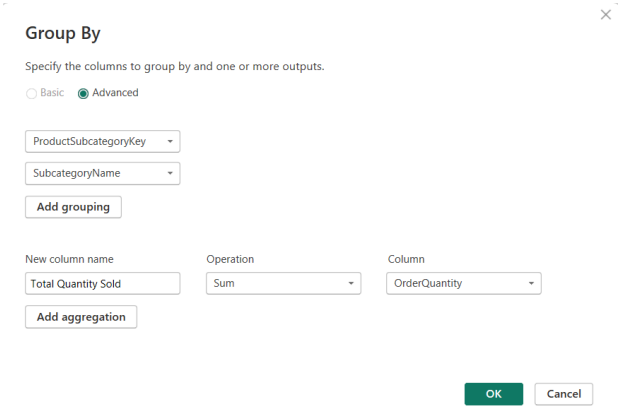


- Select top 10 rows using ‘Keep Rows’ which is on the Home tab. Here is the final result:

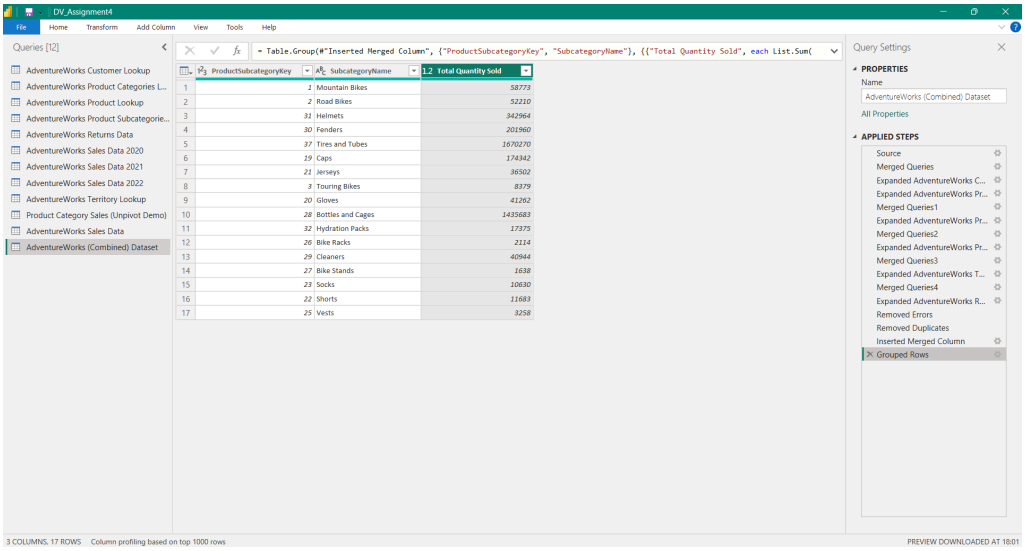


4. Find the total quantity sold for each product subcategory.

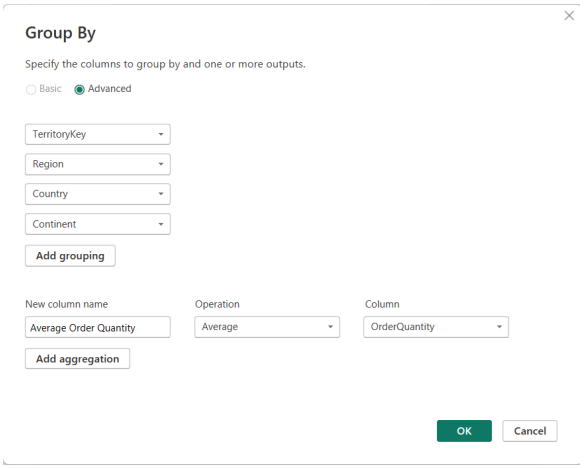
- Click on GroupBy which is on the Home / Transform tab. Go to ‘Advanced’ mode. Select ‘ProductSubcategoryKey’ and ‘SubcategoryName’ for grouping ‘Total Quantity Sold’.
- Total Quantity Sold = SUM([OrderQuantity])



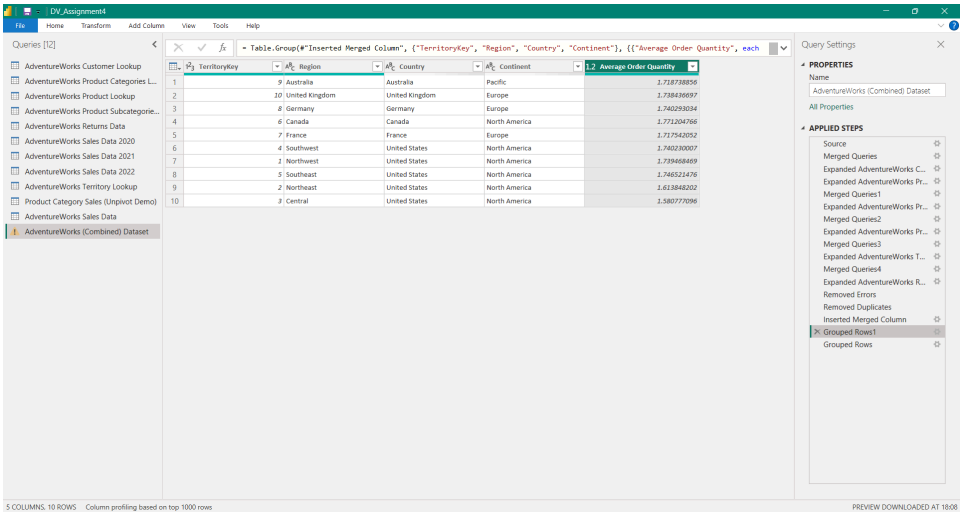
- Here is the result:



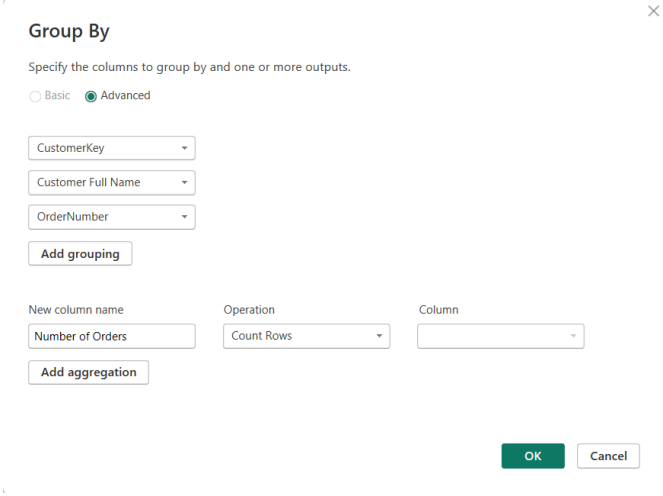
5. Determine the average order quantity for each sales territory.
   - Click on GroupBy which is on the Home / Transform tab. Go to ‘Advanced’ mode. Select ‘TerritoryKey’, ‘Region’, ‘Country; and ‘Continent’ for grouping ‘Average Order Quantity’.
   - Average Order Quantity = AVERAGE([OrderQuantity])



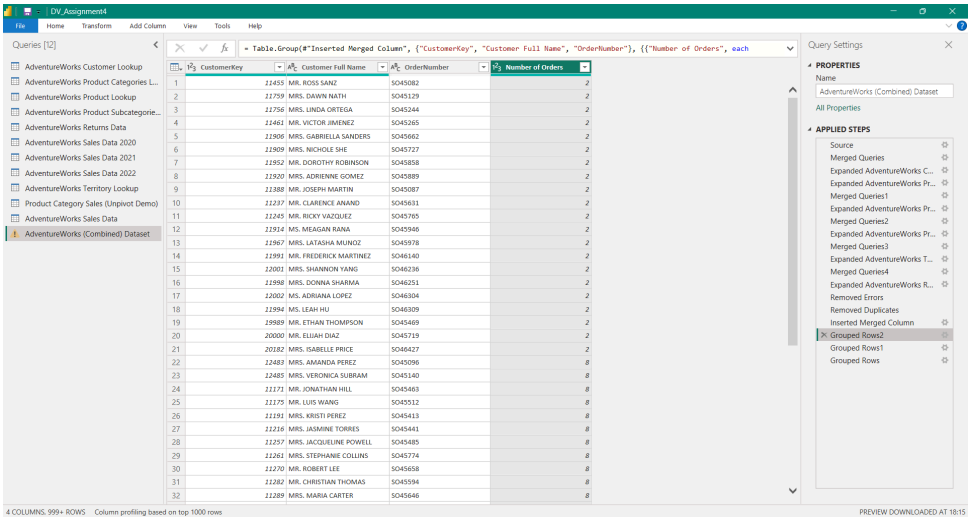
- Here is the result:



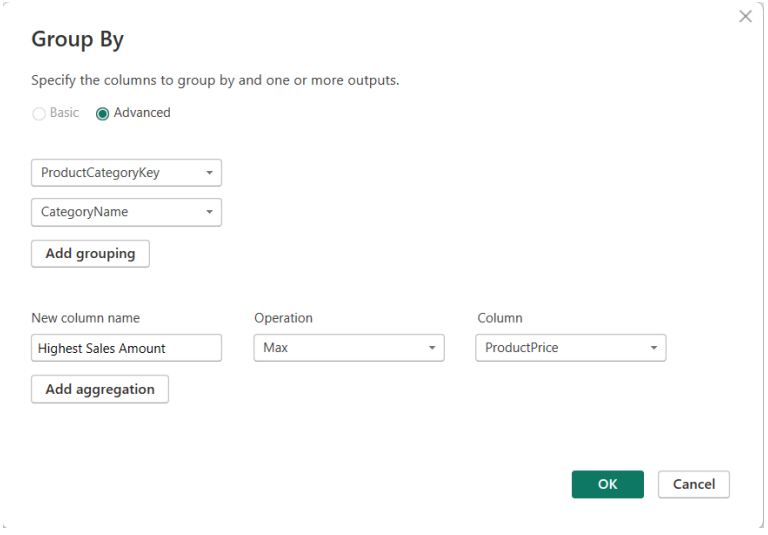
6. Count the number of orders placed by each customer.
    - Click on GroupBy which is on the Home / Transform tab. Go to ‘Advanced’ mode. Select ‘CustomerKey’, ‘Customer Full Name’ and ‘OrderNumber’ for grouping ‘Number of Orders’.
    - Number of Orders = COUNTROWS



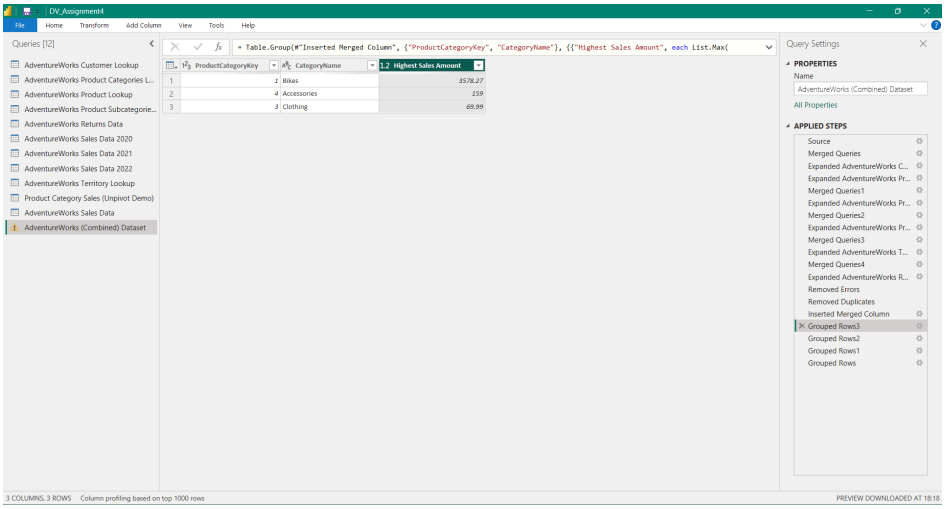
- Here is the result:



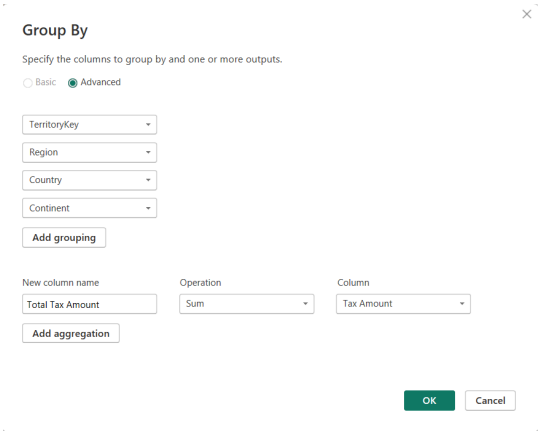
7. Find the highest sales amount recorded for each product category.
    - Click on GroupBy which is on the Home / Transform tab. Go to ‘Advanced’ mode. Select ‘ProductCategoryKey’ and ‘CategoryName’ for grouping ‘Highest Sales Amount’.
    - Highest Sales Amount = MAX([ProductPrice])



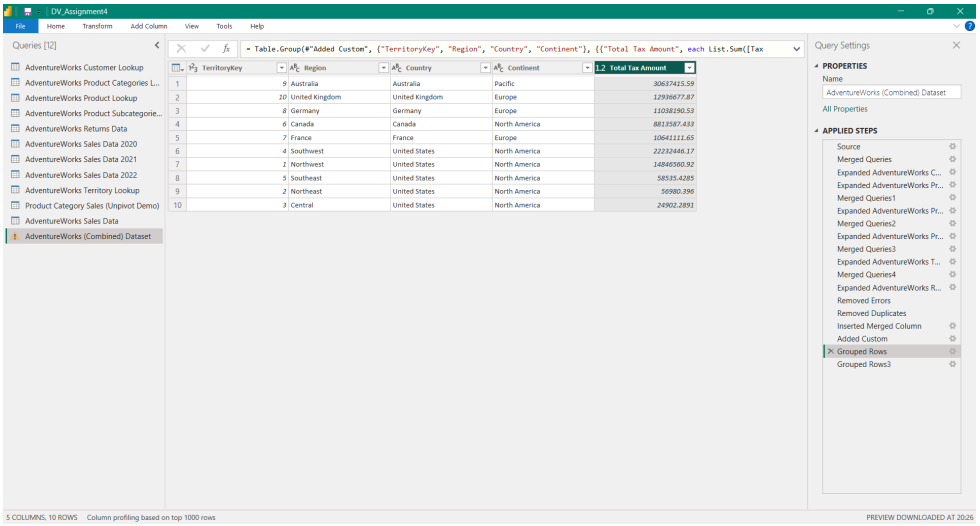
- Here is the result:



8. Calculate the total tax amount collected for each sales territory.
    - Click on GroupBy which is on the Home / Transform tab. Go to ‘Advanced’ mode. Select ‘TerritoryKey’, ‘Region’, ‘Country; and ‘Continent’ for grouping ‘Total Tax Amount’.
    - Total Tax Amount = SUM([Tax Amount])

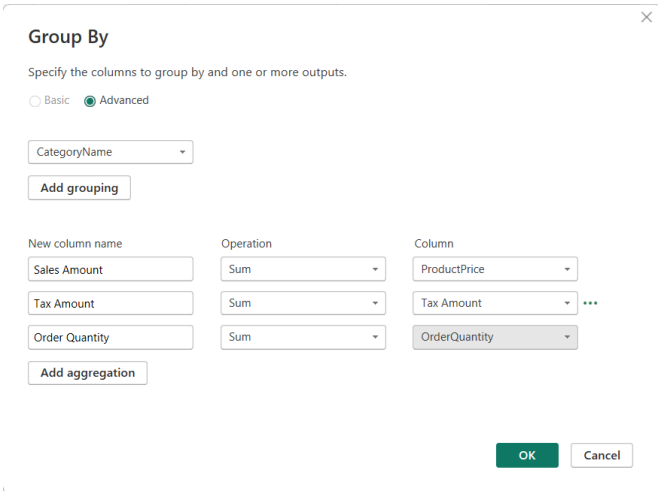


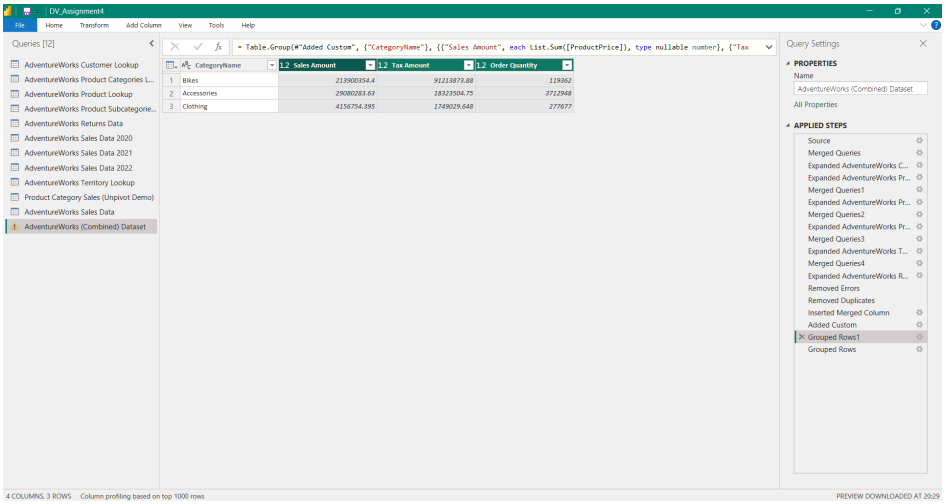
- Here is the result:



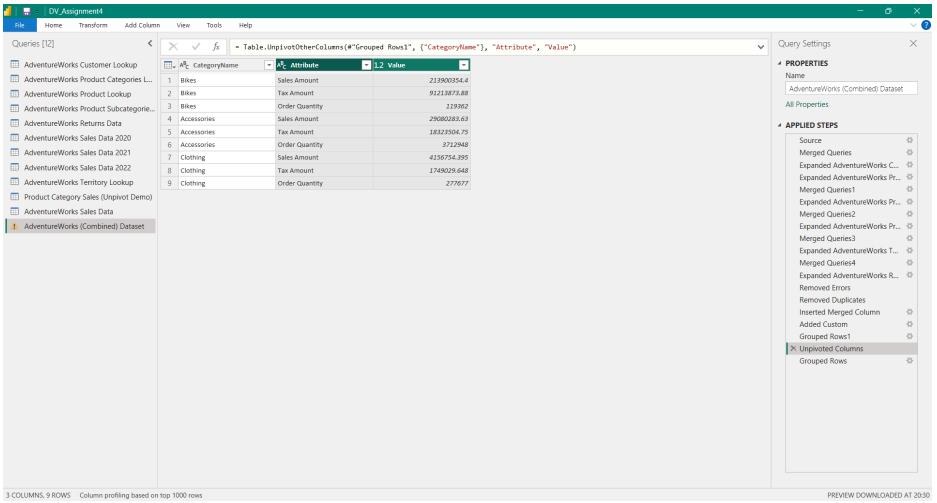
B. Pivoting and Unpivoting

1. Created a table containing Product Category, Sales Amount, Tax Amount, and Order Quantity using GroupBy (located on Home / Transform tab).
    - Select ‘CategoryName’ for grouping ‘Sales Amount’, ‘Tax Amount’, and ‘OrderQuantity’.
    - Sales Amount = SUM([ProductPrice])
    - Tax Amount = SUM([Tax Amount])
    - Order Quantity = SUM([OrderQuantity])

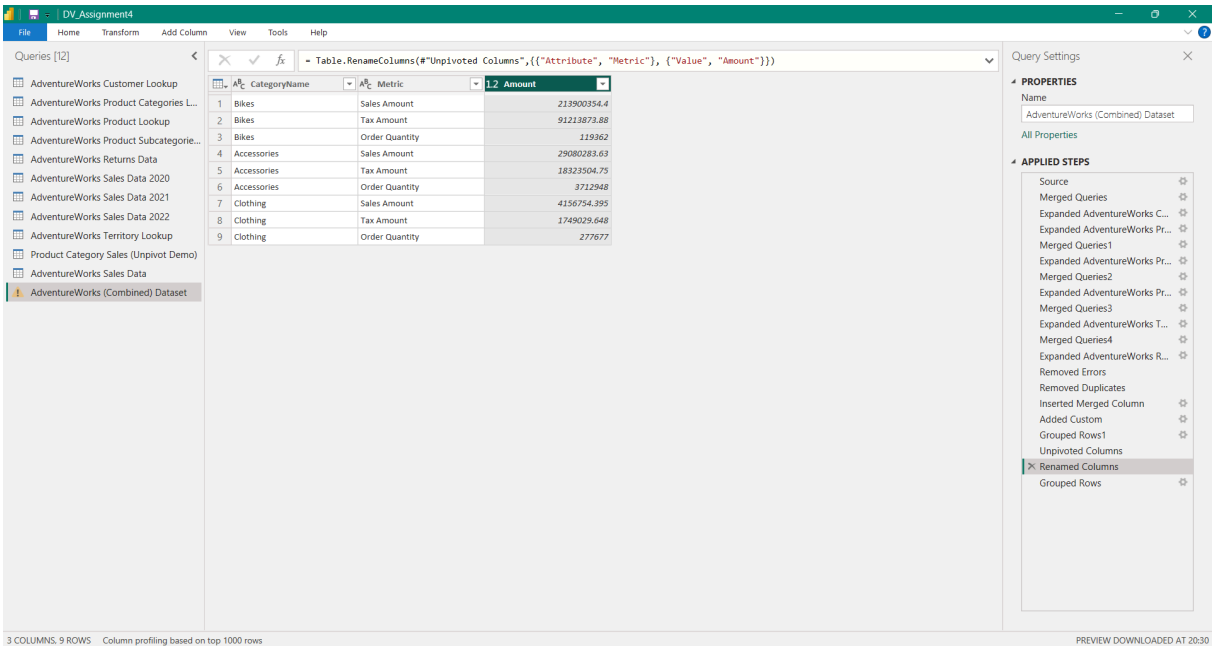




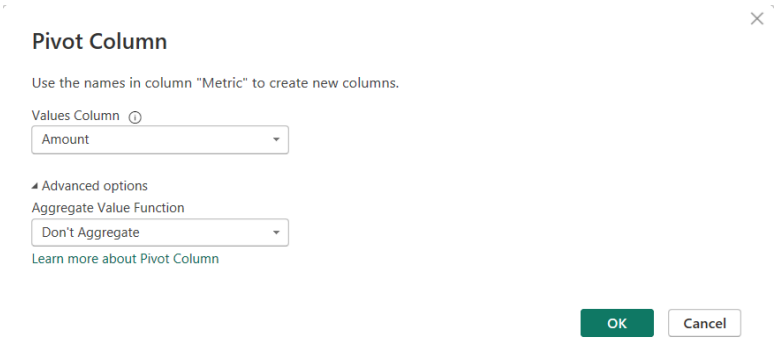
2. Unpivot the numerical columns (Sales Amount, Tax Amount, Order Quantity) into Attribute and Value columns.

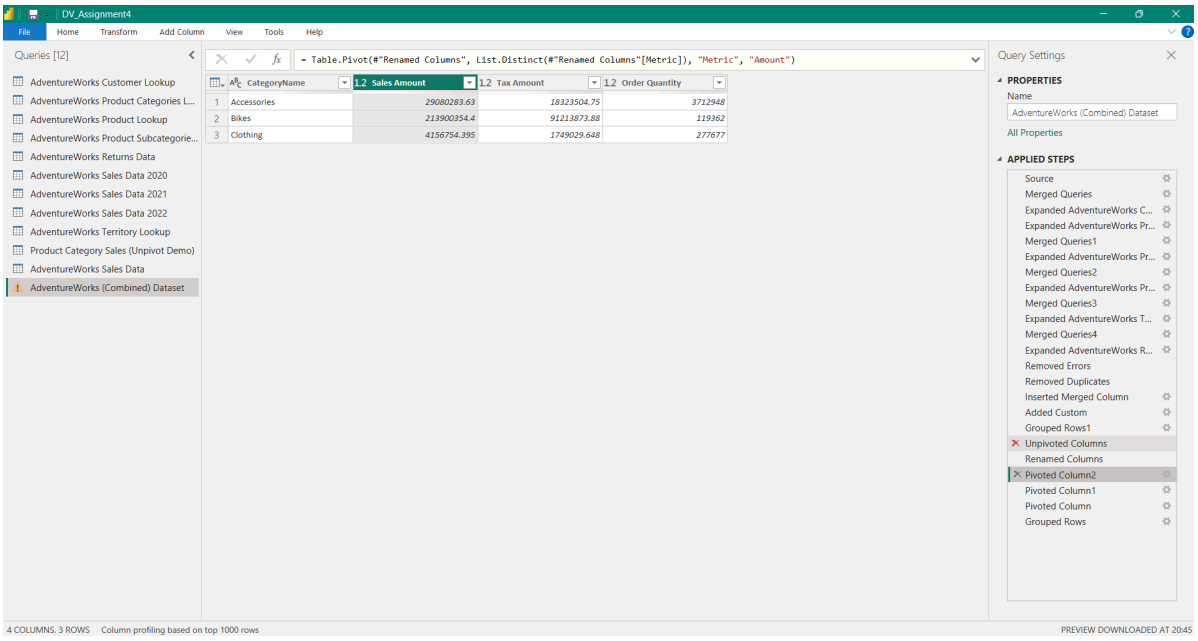


3. Rename the columns as follows:
    - Attribute → Metric
    - Value → Amount



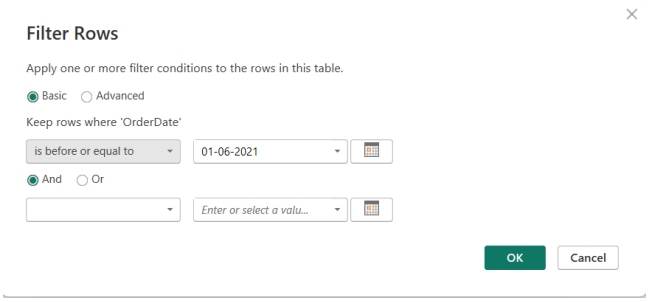
4. Pivot the Metric column back to its original structure.
    - Select the “Metric” column.
    - Click on “Pivot” and choose “Amount” in the Values Column and select “Don’t aggregate”.

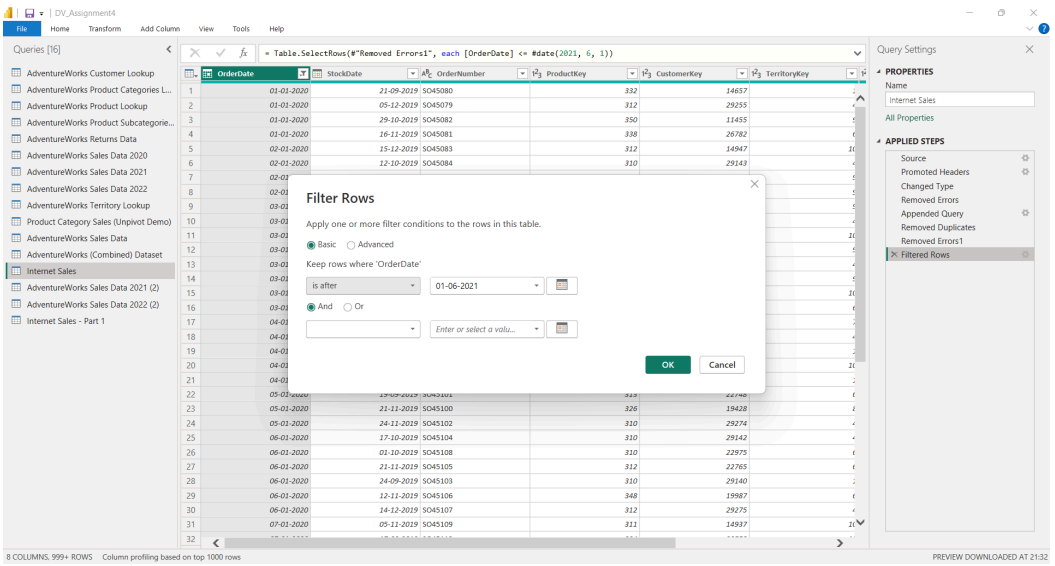


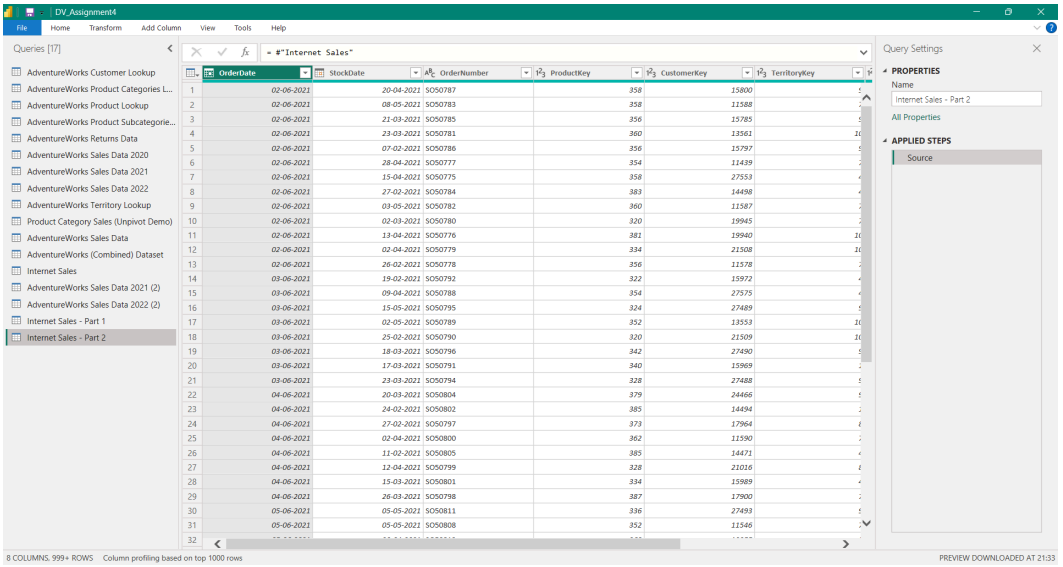


C. Append Queries

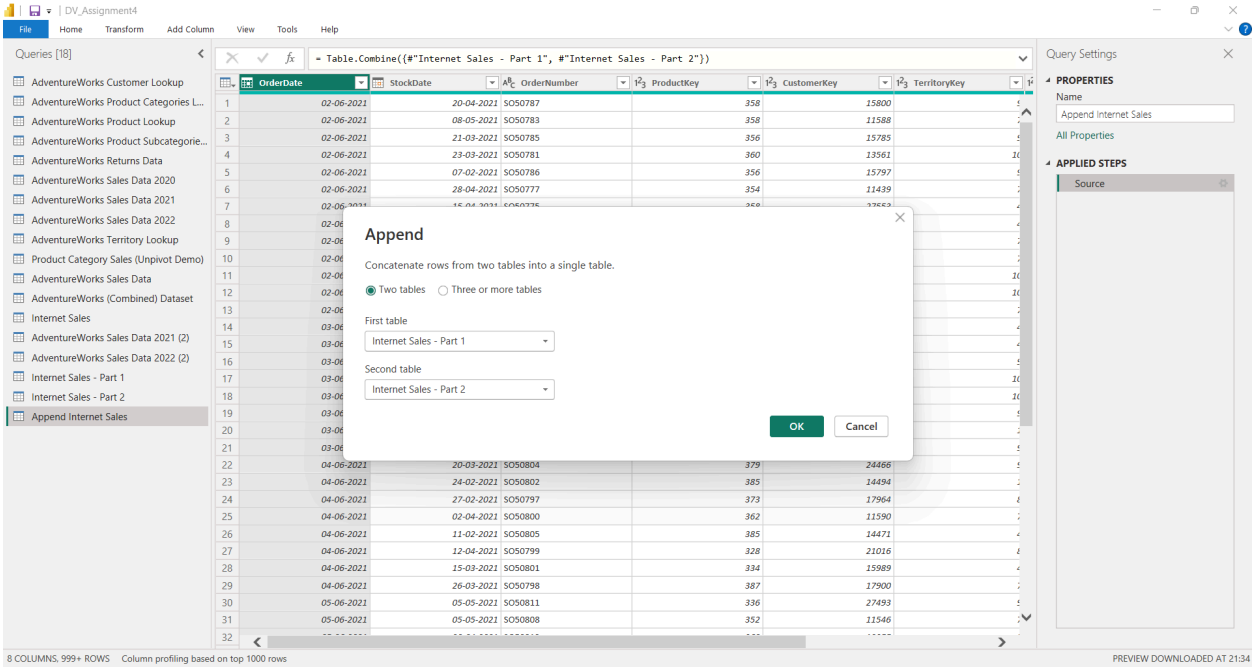
1. Create two queries:
    - Internet Sales (First Half)
    - Internet Sales (Second Half)
    - Table “Internet Sales - Part 1” was made by using ‘Filter’ by clicking on the ‘OrderDate’ column. I selected ‘before or equal to’ ‘01/06/2021” and clicked on reference to create it separately and renamed it.
    - Table “Internet Sales - Part 2” was made by using ‘Filter’ by clicking on the ‘OrderDate’ column. I selected ‘after’ ‘01/06/2021” and clicked on the reference to create it separately and renamed it.





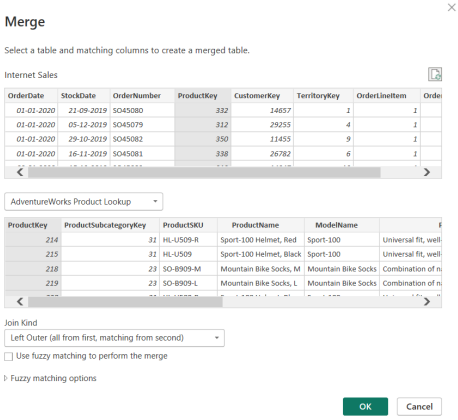


2. Append the two queries into a single table.

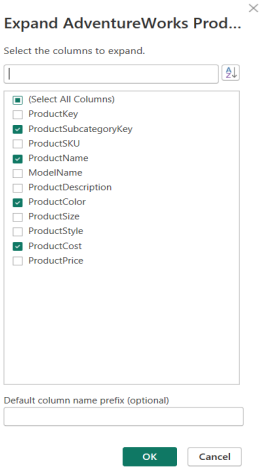


D. Merge Queries

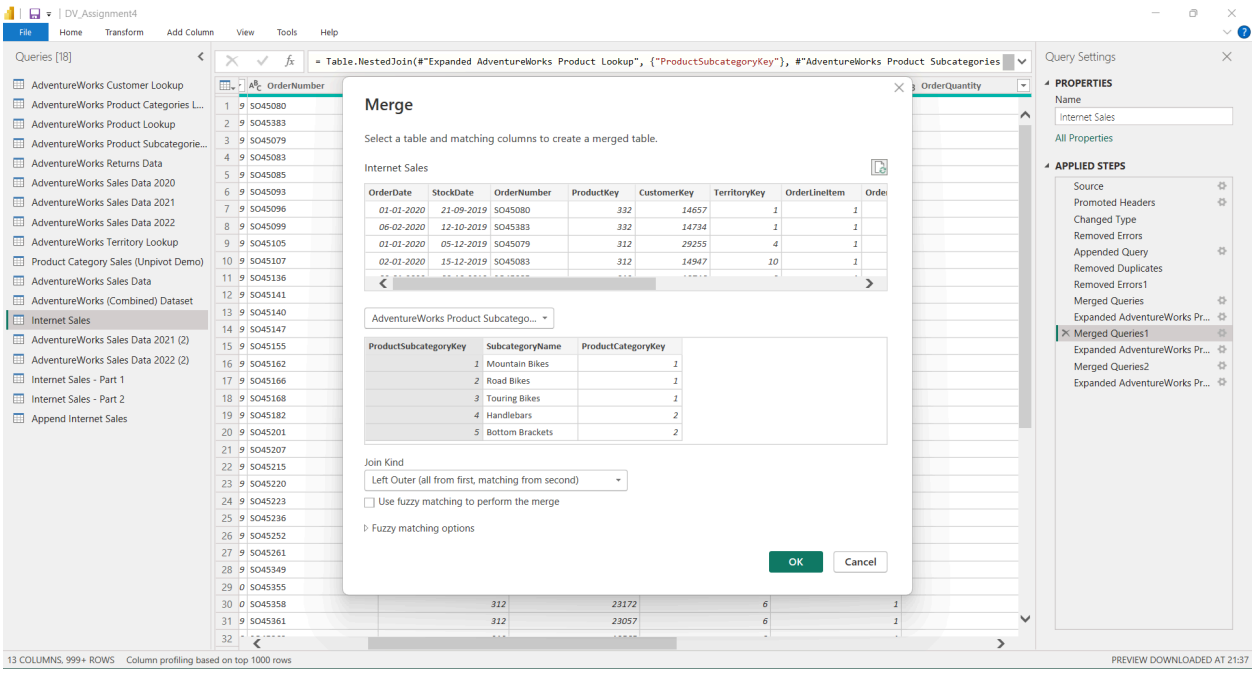
1. Merge ‘Internet Sales’ table with ‘AdventureWorks Product Lookup’ table using ProductKey.



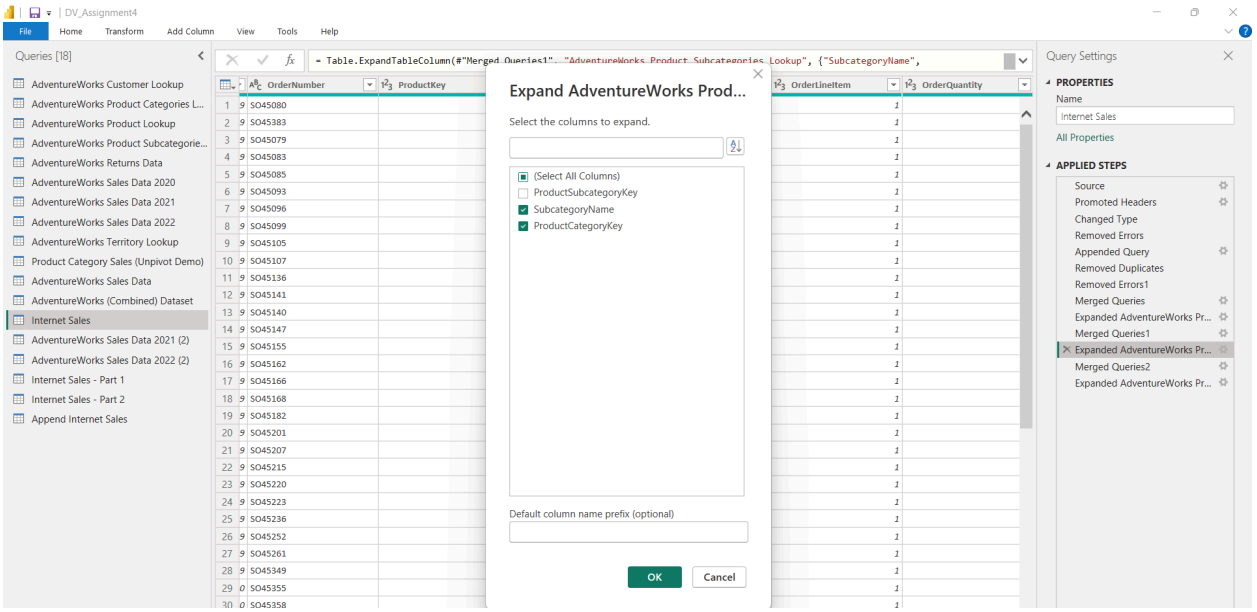
2. Expanded the merged table to include:
   - ProductName
   - ProductColor
   - ProductCost



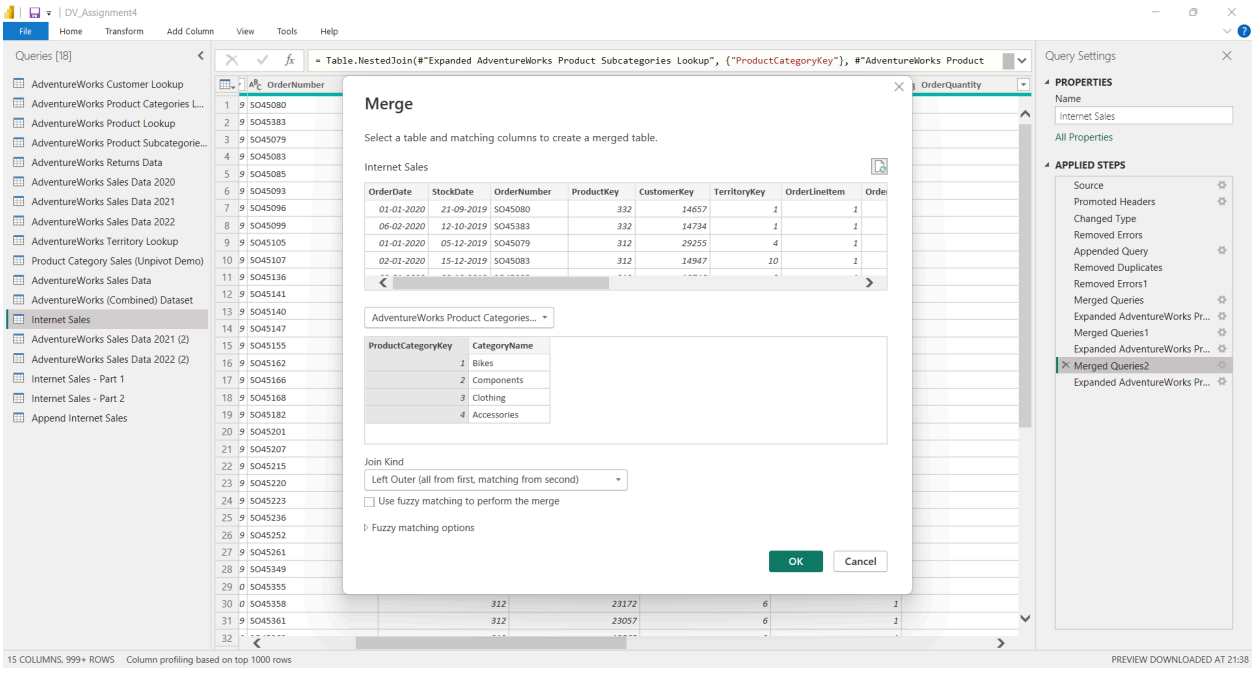
3. Merge the result with ‘AdventureWorks Product Subcategories’ table.



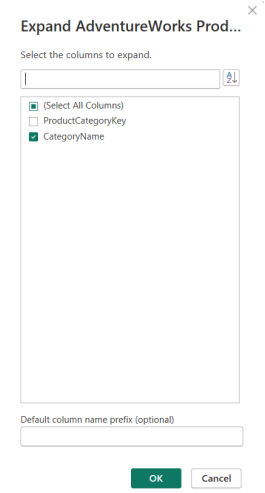
4. Expanded to include SubcategoryName and ProductCategoryKey.



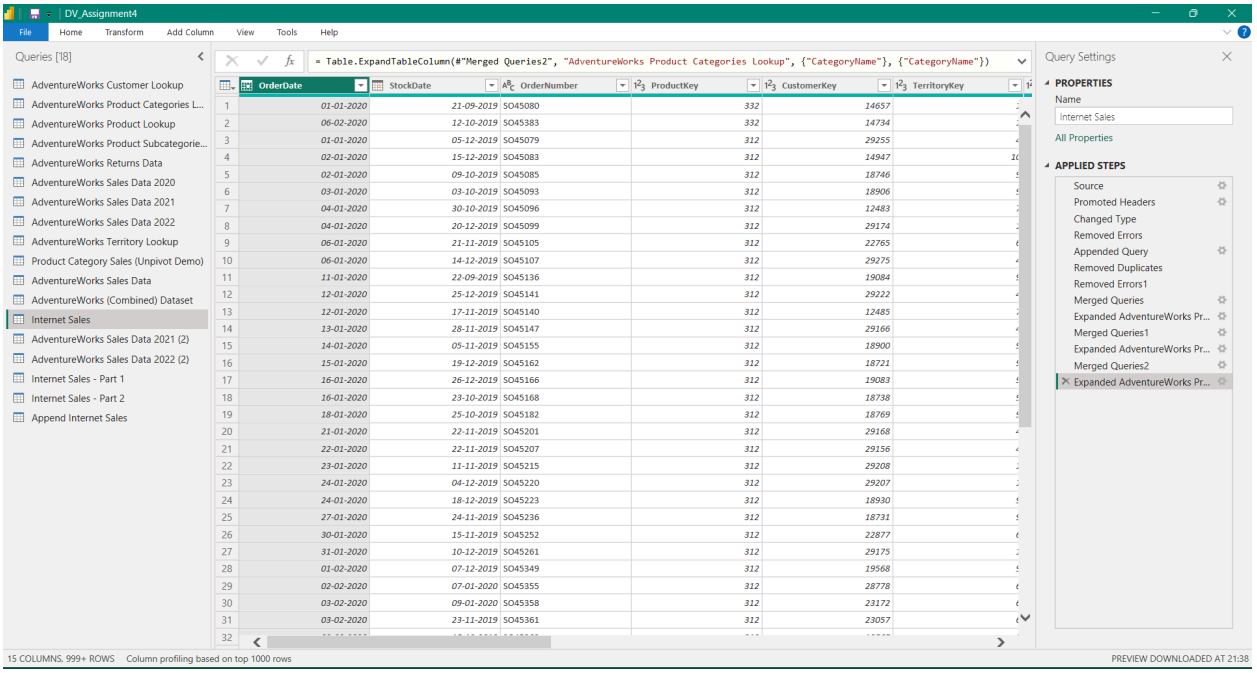
5. Merge the result with ‘AdventureWorks Product Categories’ table.

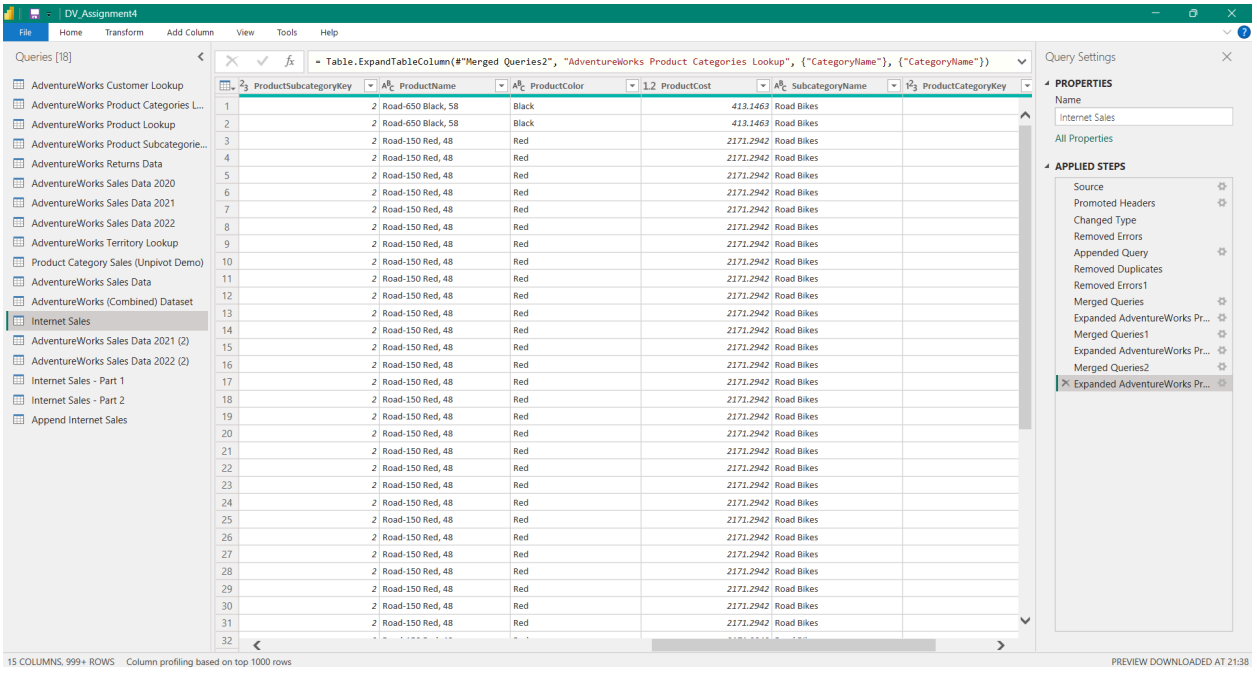


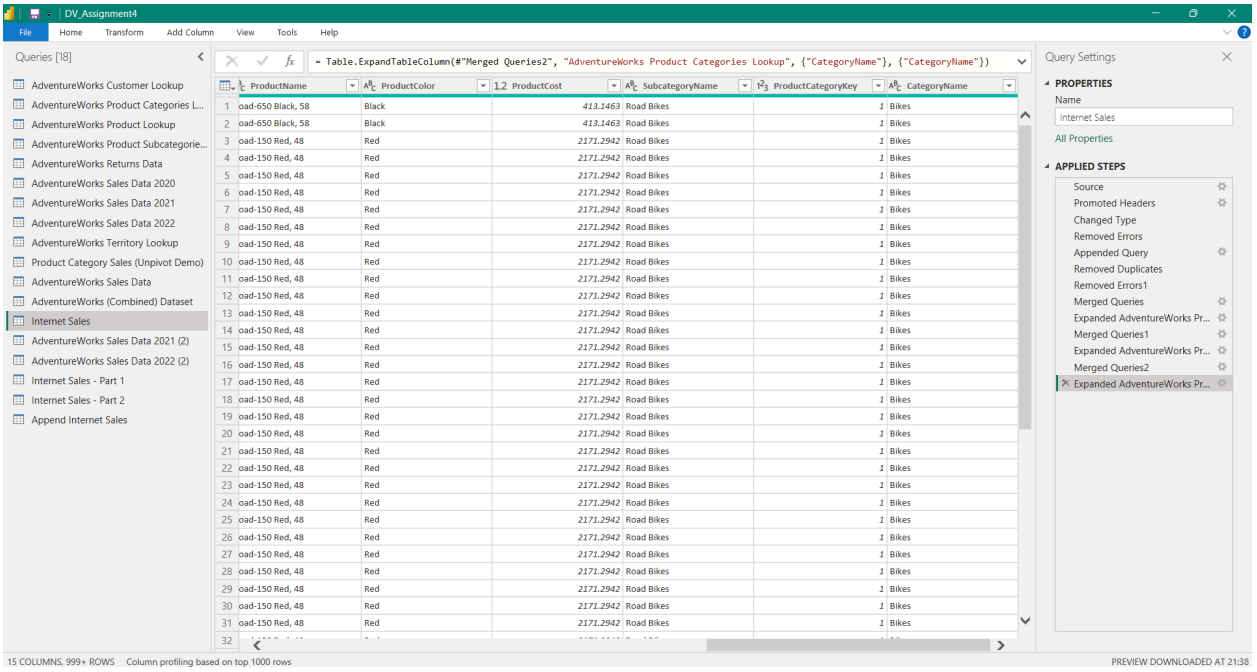
6. Expanded to include CategoryName.



7. Here's the final table:



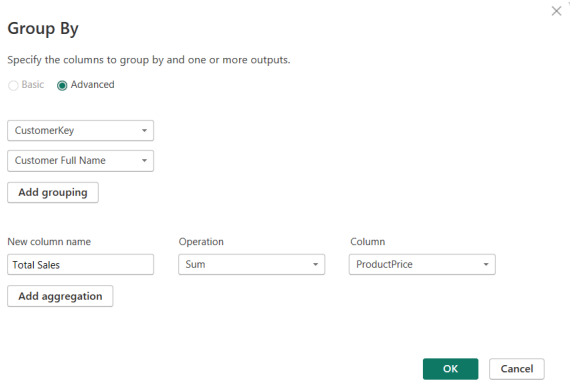




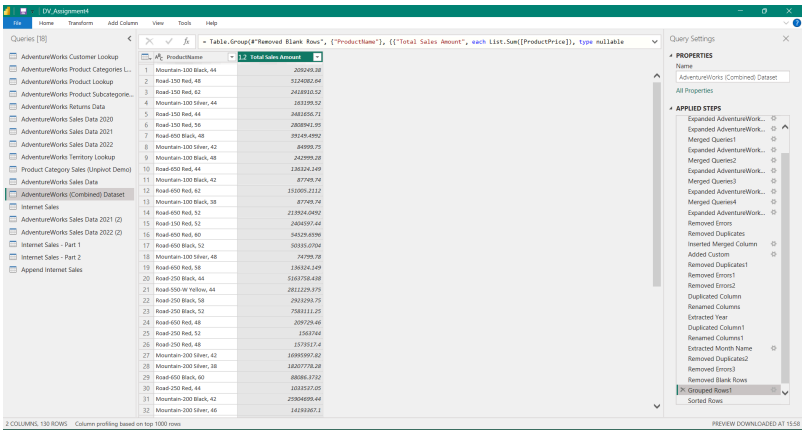
Part B: Sales Analysis

Perform the following analyses:

1. Calculate Total Sales by Product Category.
   - Click on GroupBy which is on the Home / Transform tab. Go to ‘Advanced’ mode. Select ‘CustomerKey’ and ‘Customer Full Name’ for grouping ‘Total Sales’.
   - Total Sales = SUM([ProductPrice])

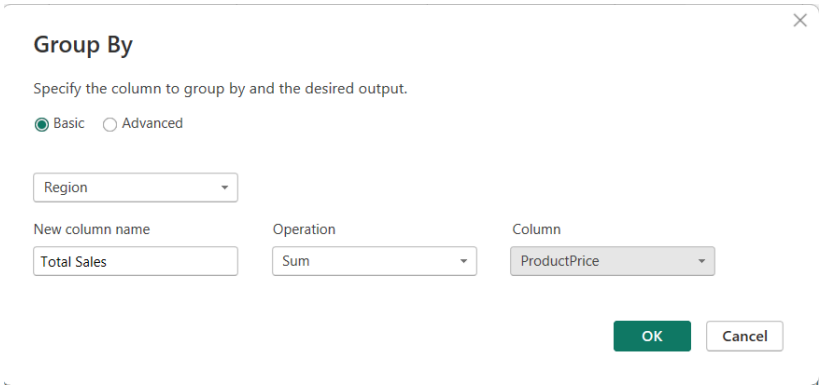


   - Here is the result:

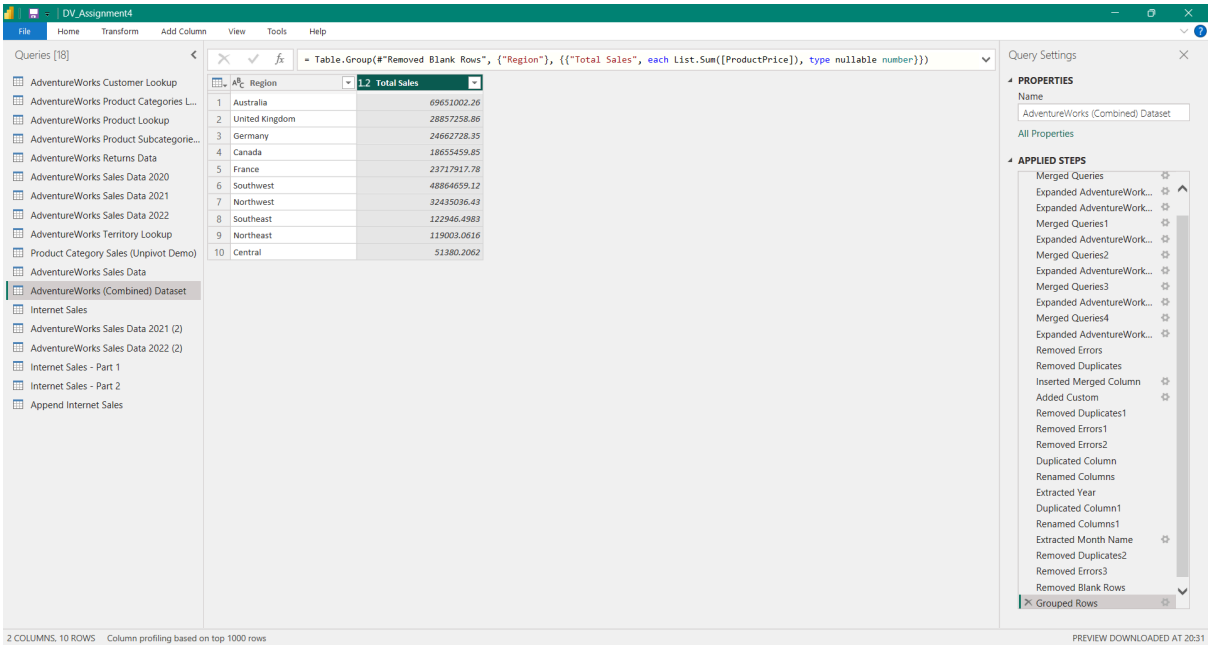


2. Calculate Total Sales by Sales Territory.

- Click on GroupBy which is on the Home / Transform tab. Select ‘ProductName’ for grouping ‘Total Sales’.
- Total Sales = SUM([ProductPrice])

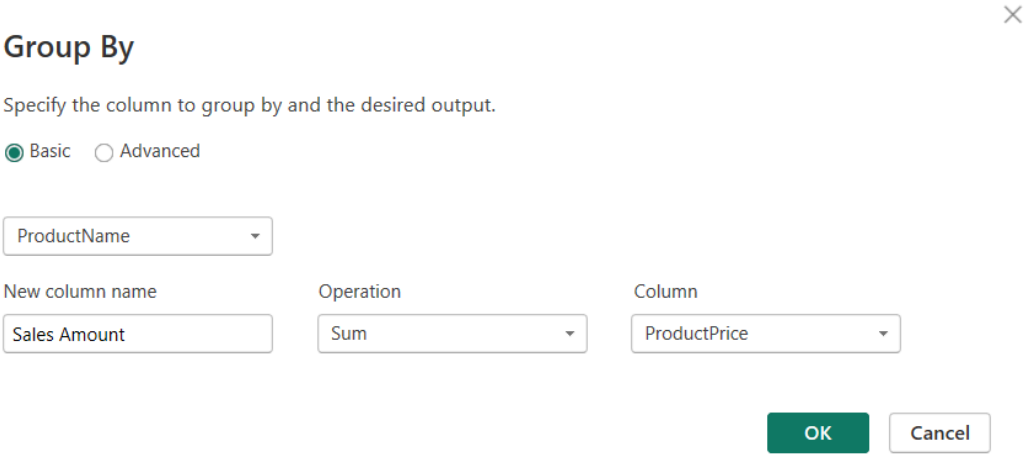


- Here is the result:

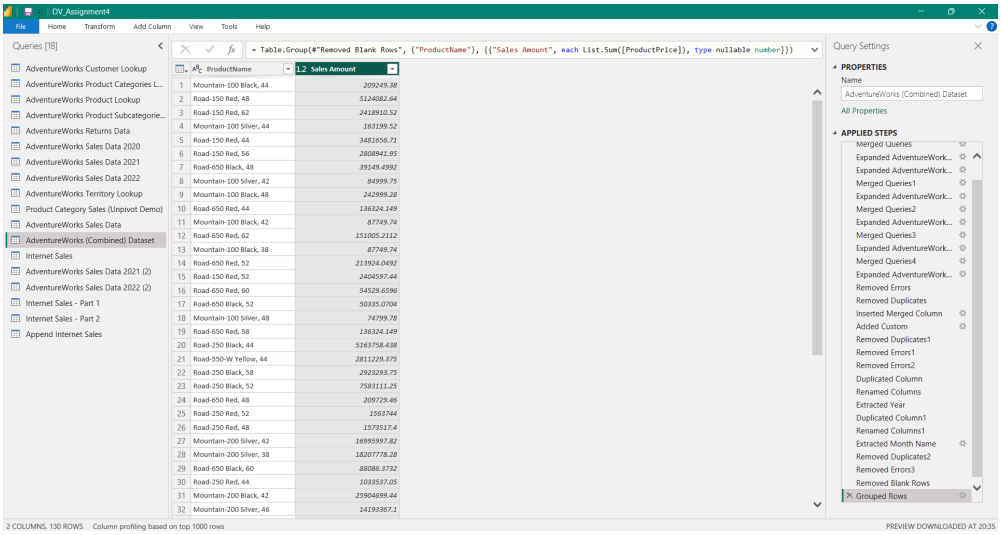


3. Find the Top 10 Products based on Sales Amount.

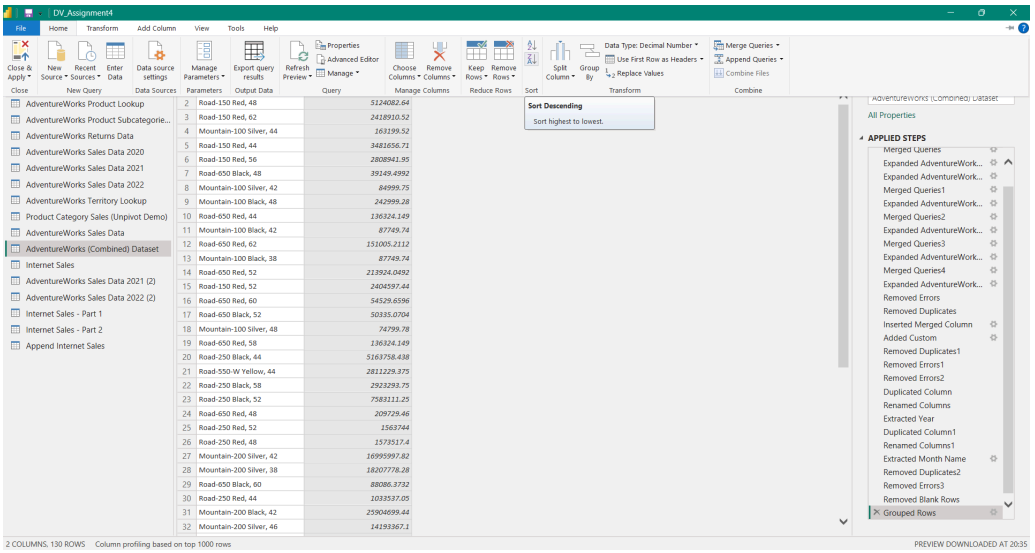
- Click on GroupBy which is on the Home / Transform tab. Select ‘ProductName’ for grouping ‘Sales Amount’.
- Sales Amount = SUM([ProductPrice])

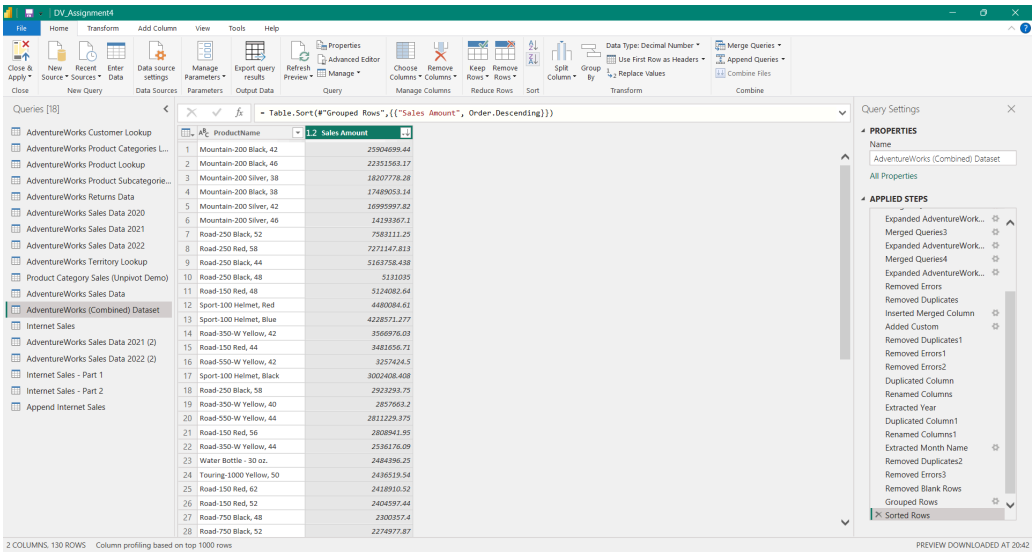


- Here is the result:



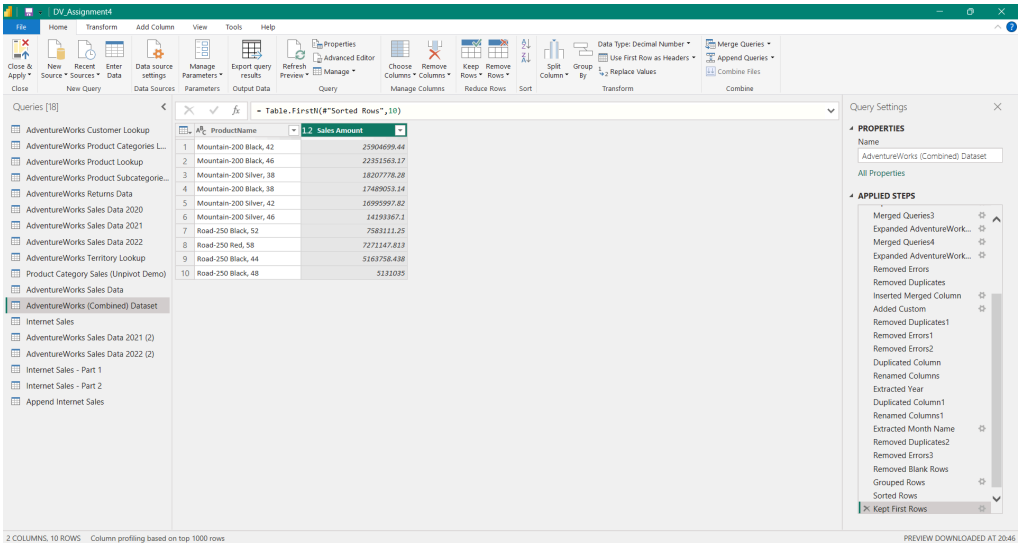
- Sort it in descending order.



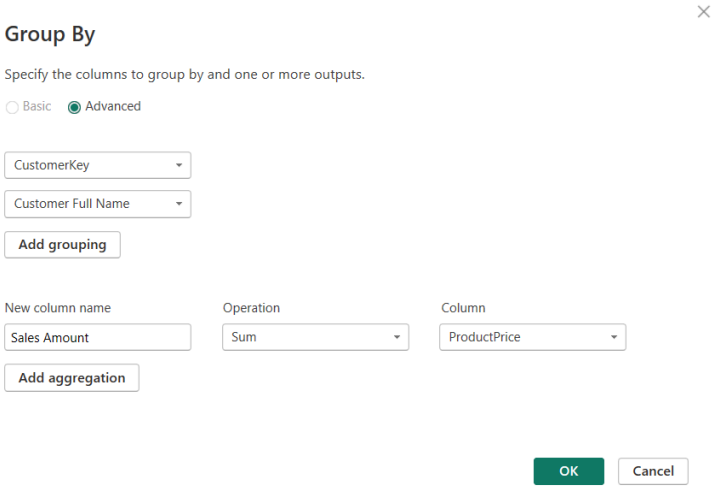


- Click on “Keep Rows” - Top 10 rows

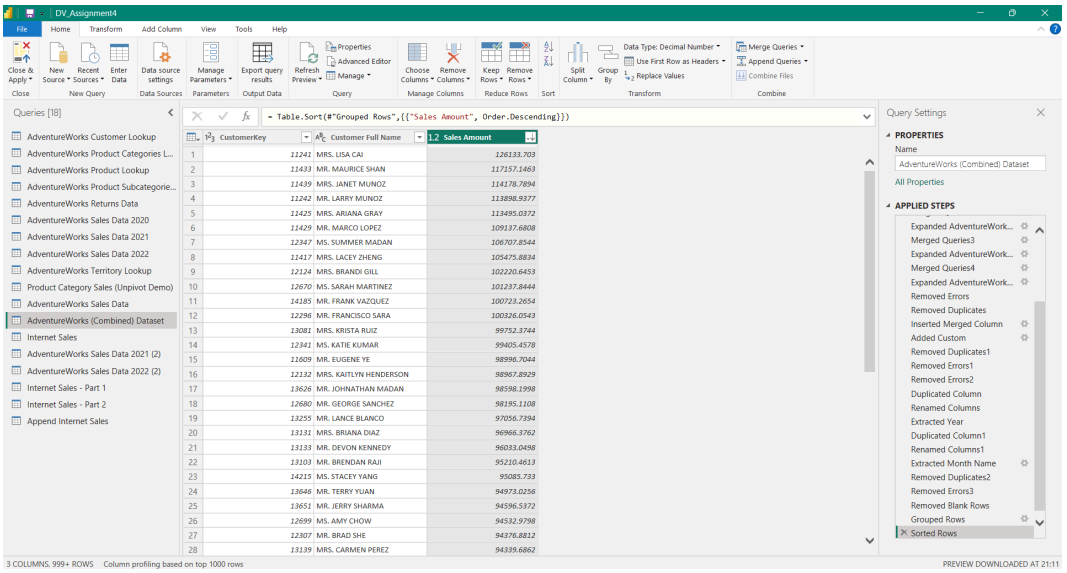




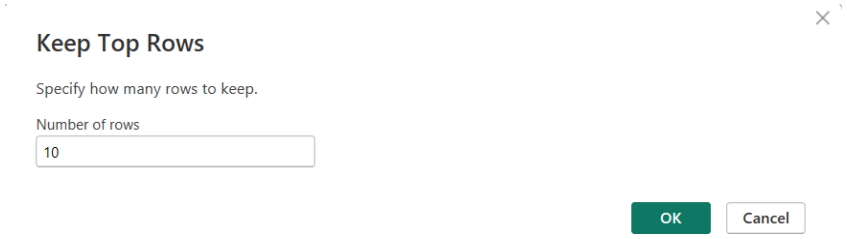
4. Determine the Top 10 Customers based on Sales Amount.
    - Click on GroupBy which is on the Home / Transform tab. Select ‘ProductName’ for grouping ‘Sales Amount’.
    - Sales Amount = SUM([ProductPrice])

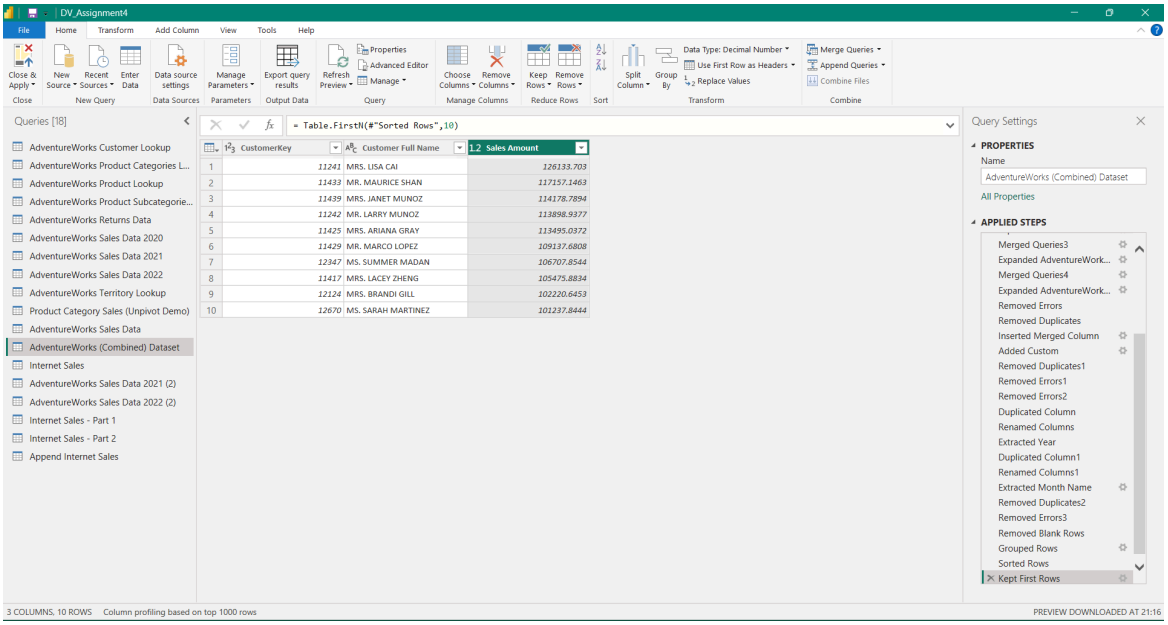


- Now, sort it in descending order by clicking on ‘Sales Amount’ and clicking on Sort (Z-A) from the Home tab.



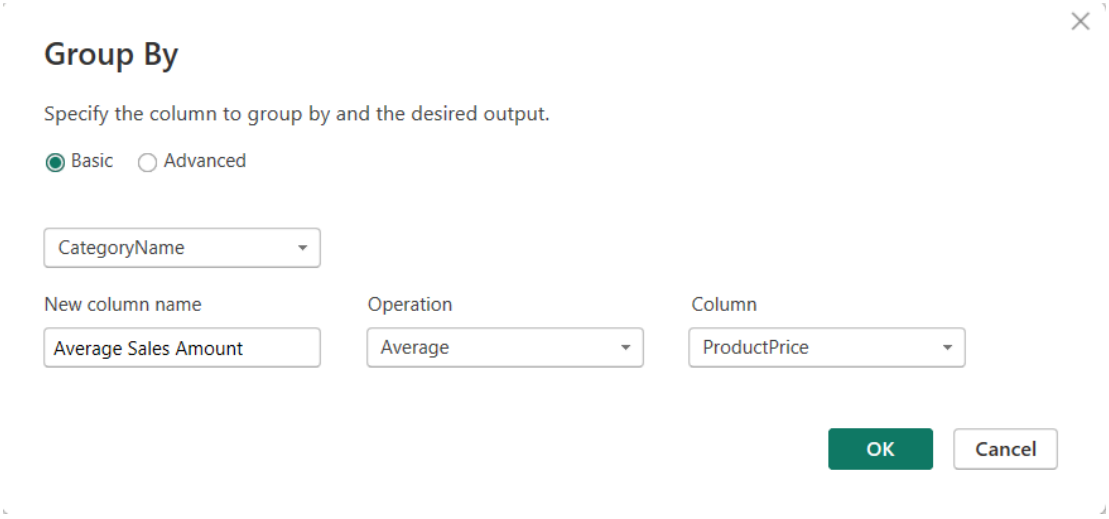
- Click on “Keep Rows” - Top 10 rows

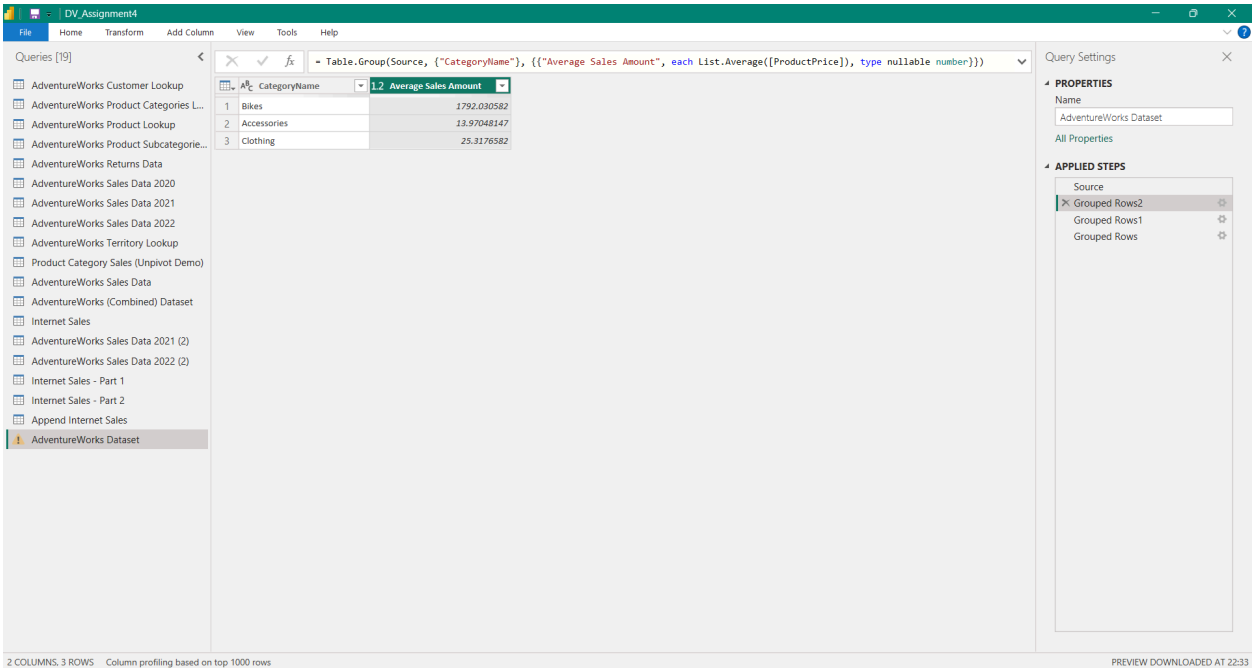




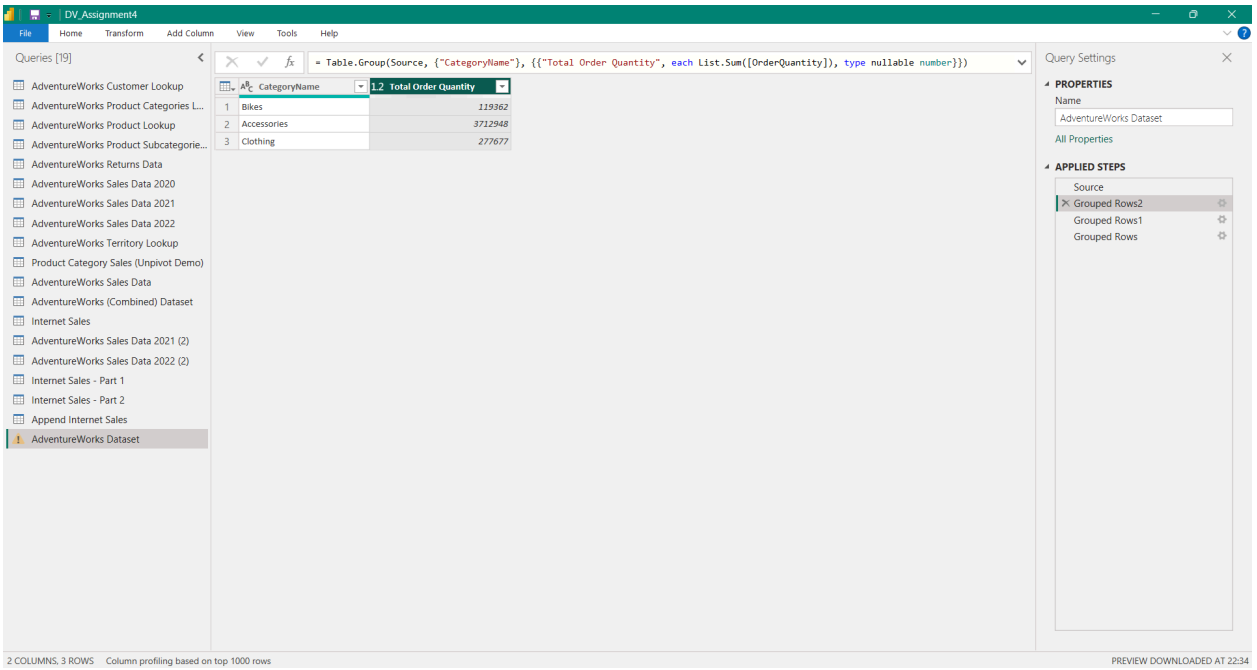
5. Calculate the Average Sales Amount by Product Category.

- Click on GroupBy which is on the Home / Transform tab. Select ‘ProductName’ for grouping ‘Average Sales Amount’.
- Average Sales Amount = AVERAGE([ProductPrice])

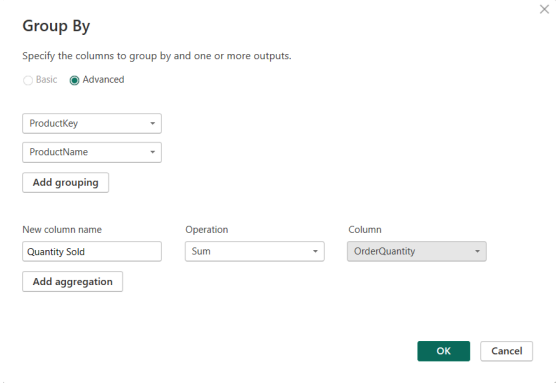


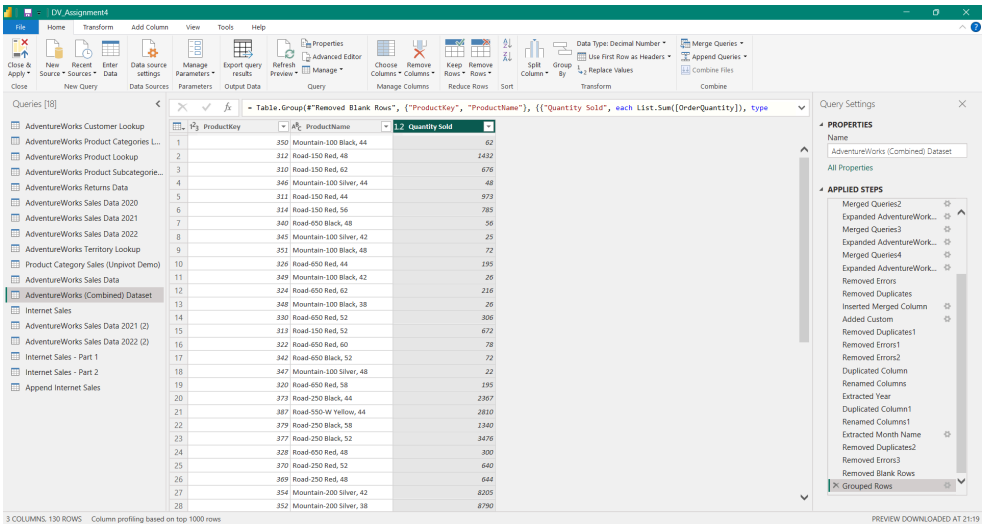


6. Find the Total Order Quantity for each Product Category.
    - Click on GroupBy which is on the Home / Transform tab. Select ‘CategoryName’ for grouping ‘Total Order Quantity’.
    - Total Order Quantity = SUM([OrderQuantity])

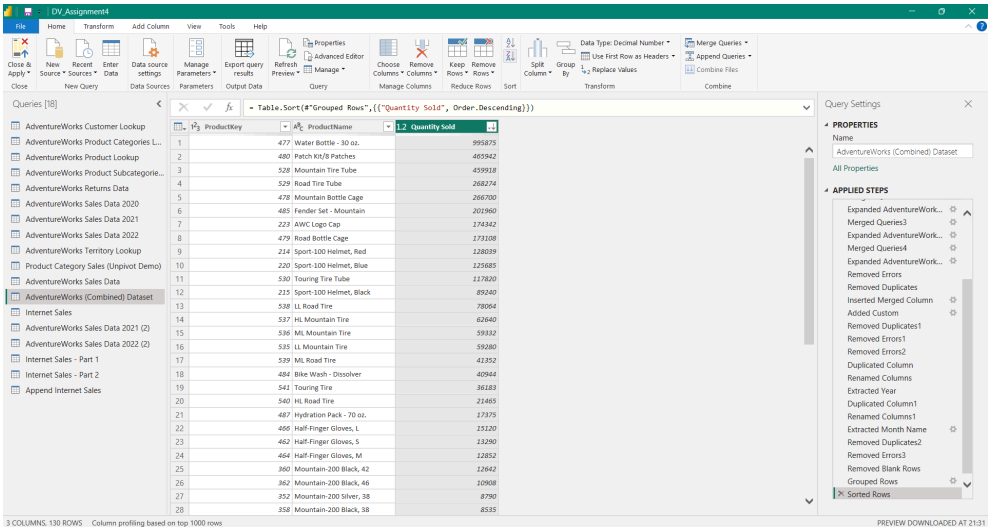


7. Determine the highest-selling product.
    - Click on GroupBy which is on the Home / Transform tab. Go to ‘Advanced’ mode. Select ‘ProductKey’ and ‘ProductFull Name’ for grouping ‘Quantity Sold’.
    - Quantity Sold = SUM([OrderQuantity])

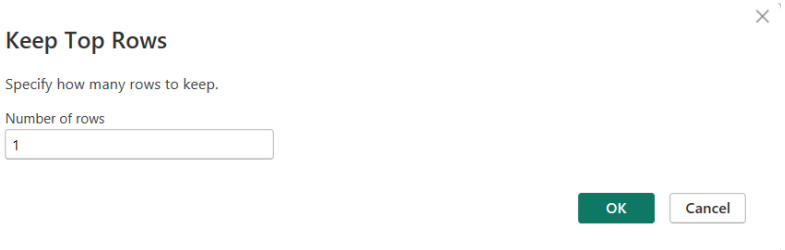


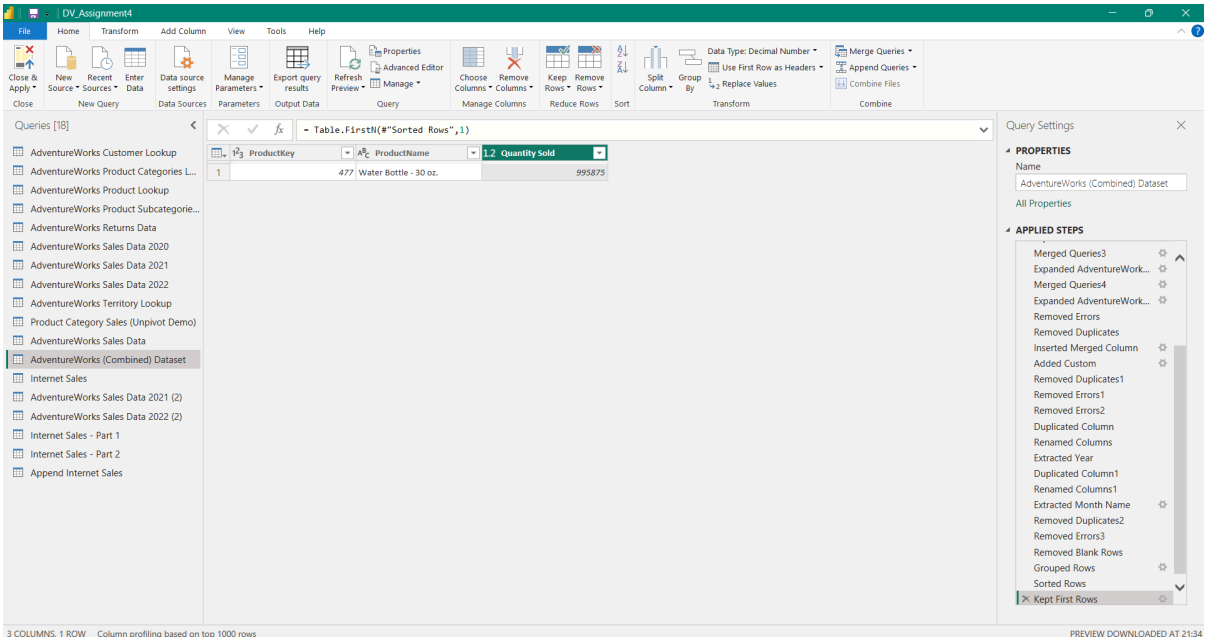


- Now, sort it in descending order by clicking on ‘Sales Amount’ and clicking on Sort (Z-A) from the Home tab.



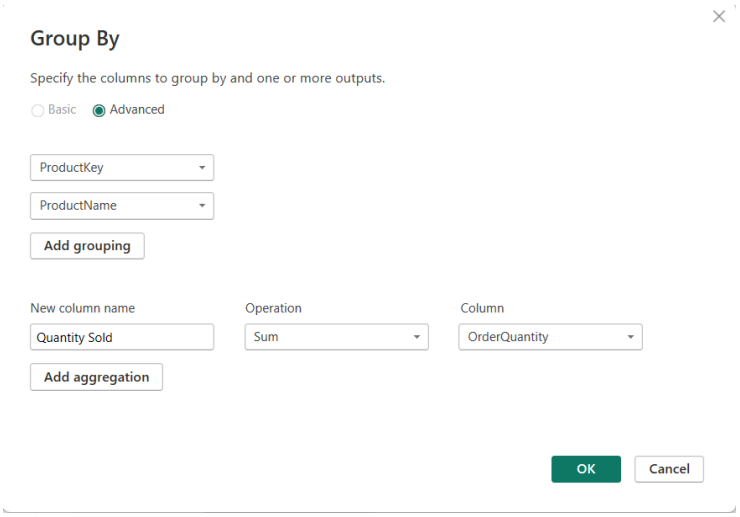
- Click on “Keep Rows” - Top 1 row

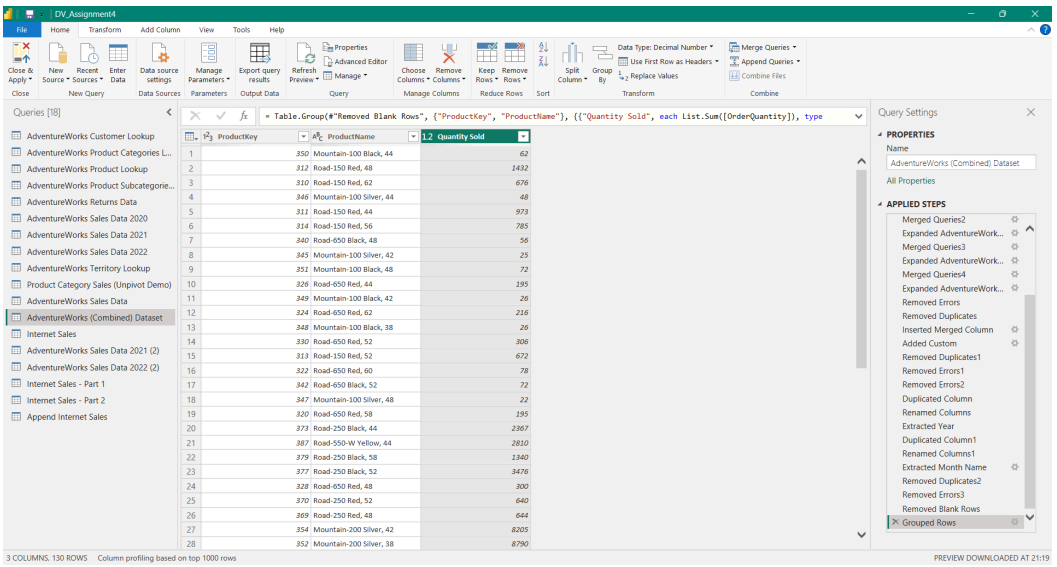




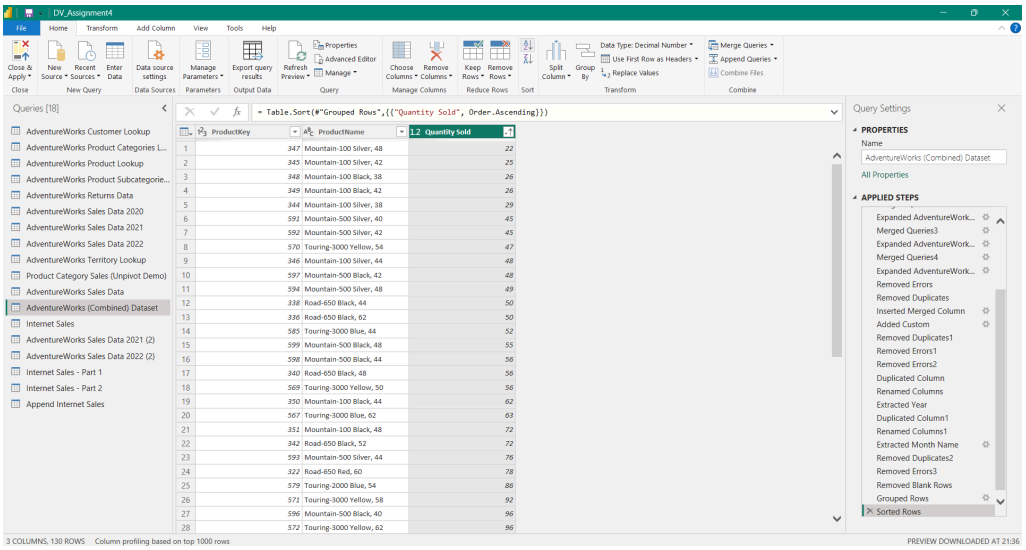
8. Identify the lowest-selling product.

- Click on GroupBy which is on the Home / Transform tab. Go to ‘Advanced’ mode. Select ‘ProductKey’ and ‘ProductFull Name’ for grouping ‘Quantity Sold’.
- Quantity Sold = SUM([OrderQuantity])

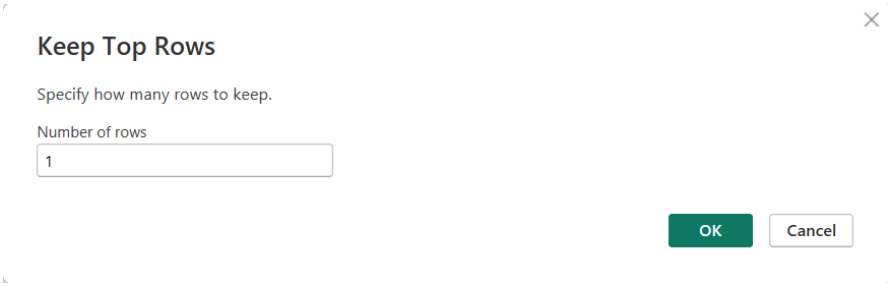


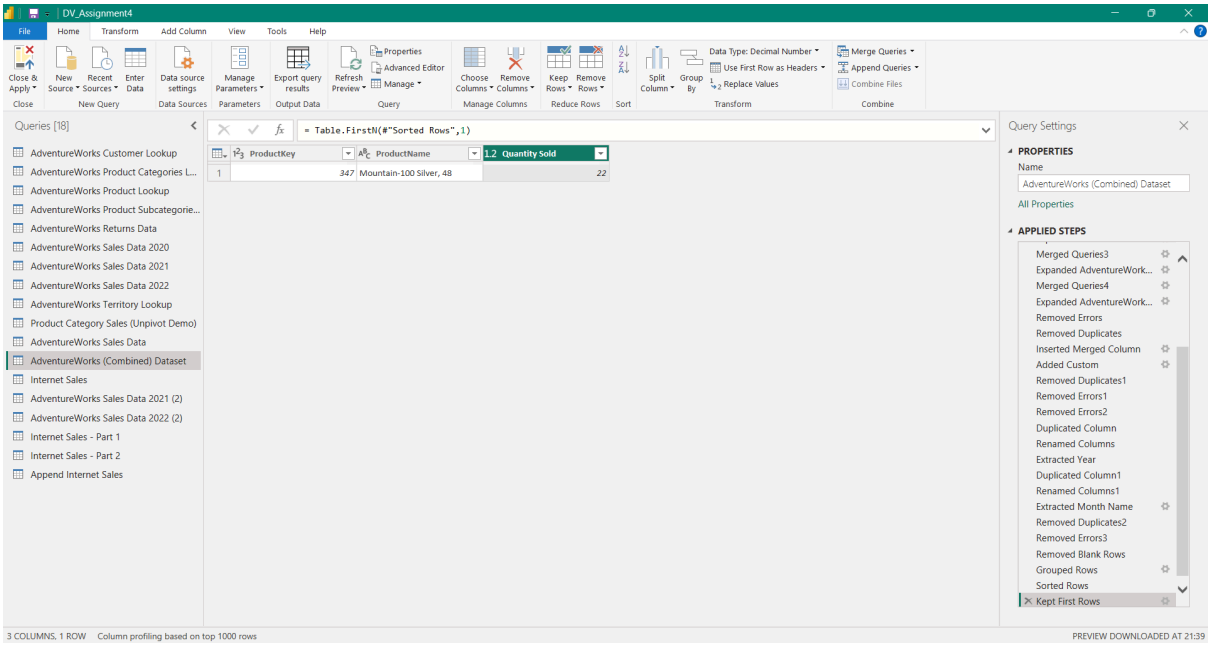


- Now, sort it in ascending order by clicking on ‘Sales Amount’ and clicking on Sort (A-Z) from the Home tab.

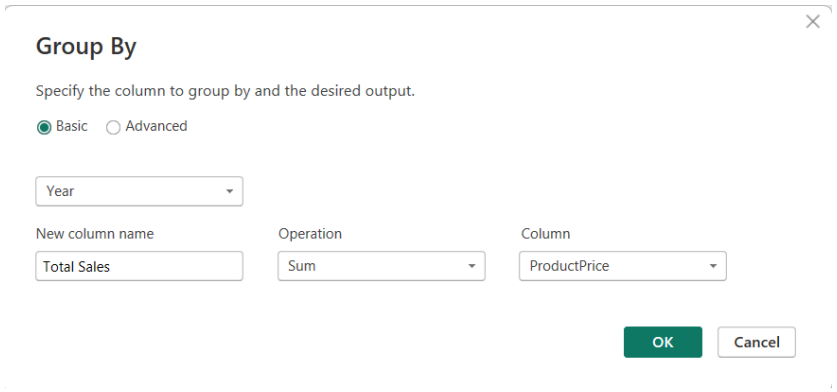


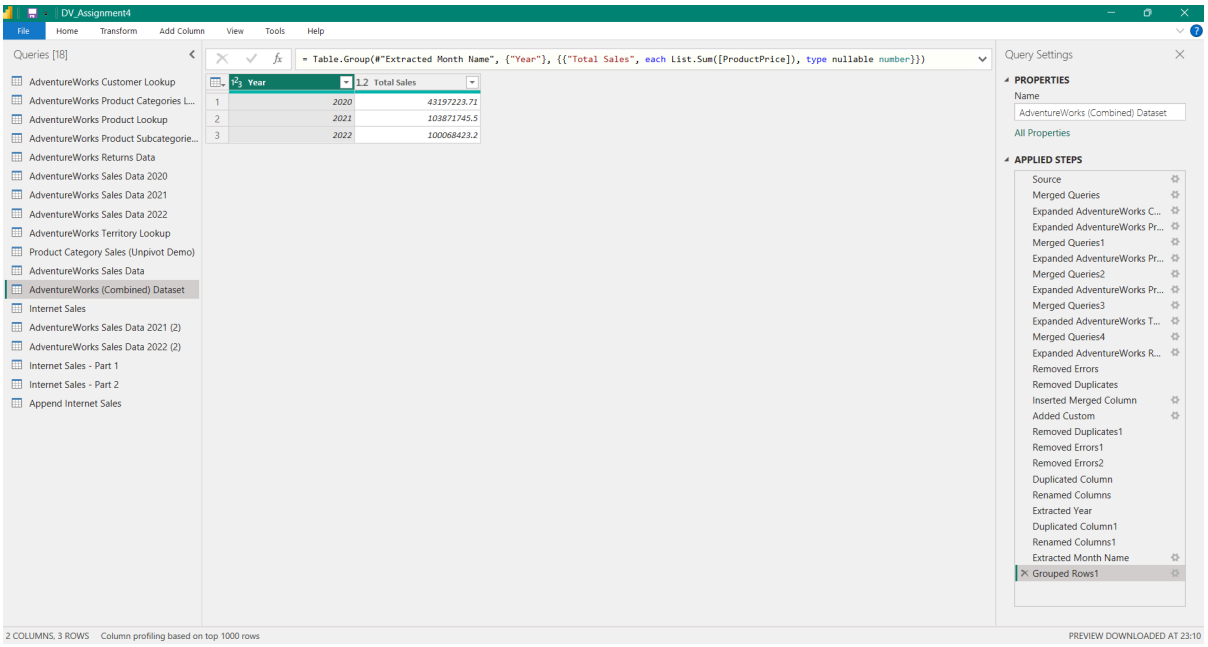
- Click on “Keep Rows” - Top 1 row



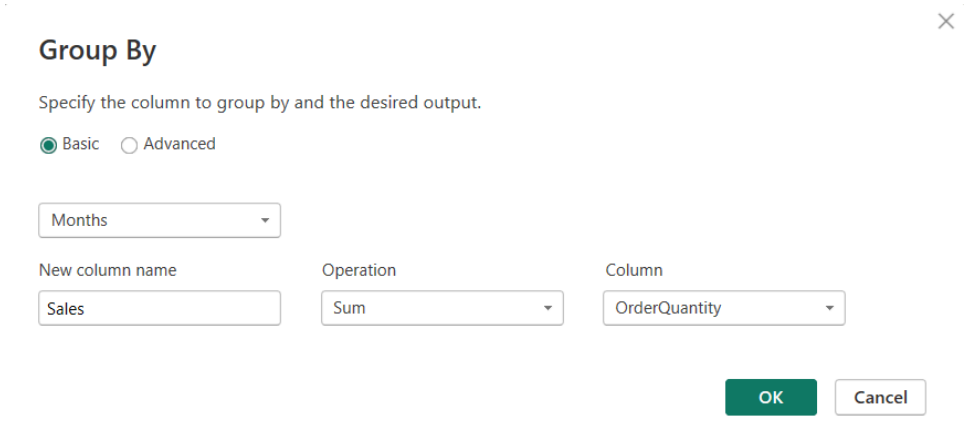


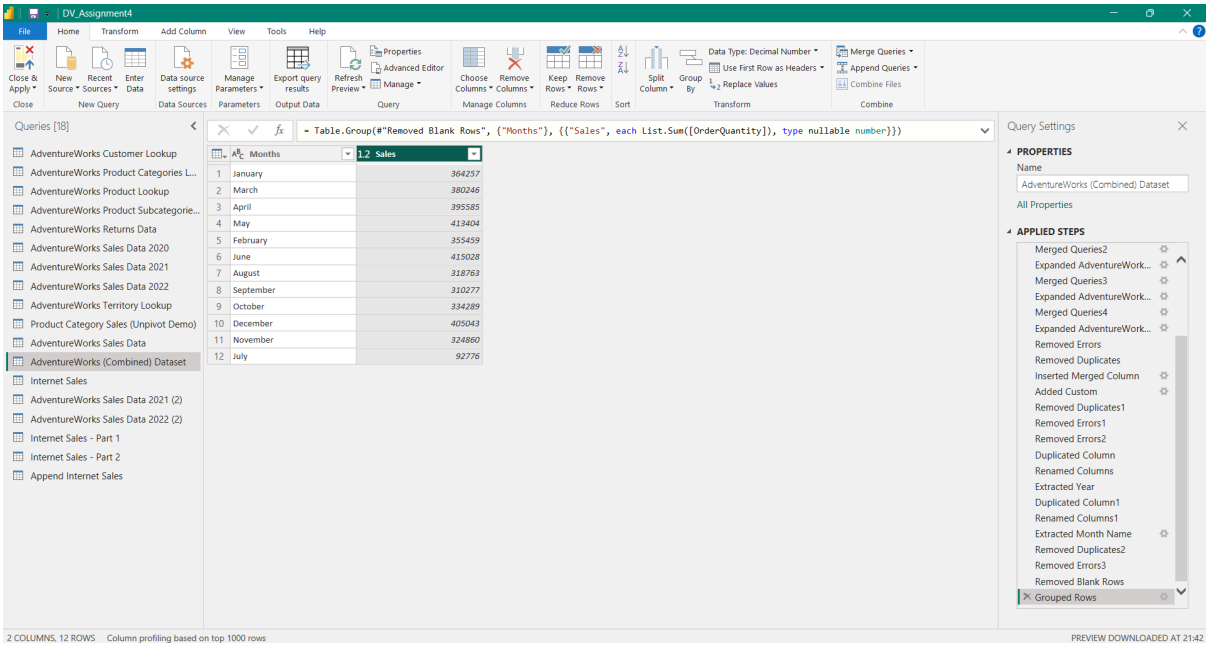
9. Calculate the total sales for each calendar year.
    - Click on GroupBy which is on the Home / Transform tab. Select ‘Year’ for grouping ‘Total Sales’.
    - Total Sales = SUM([ProductPrice])





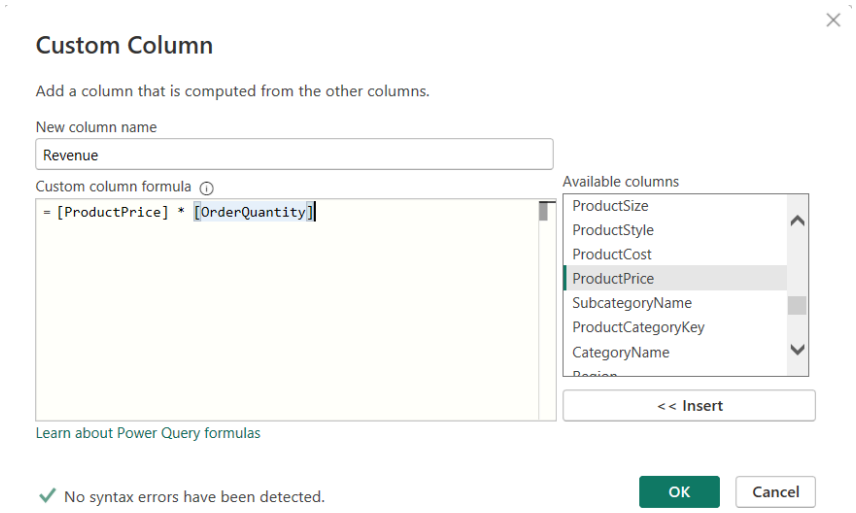
10. Compare sales across different months.
    - Click on GroupBy which is on the Home / Transform tab. Select ‘Months’ for grouping ‘Sales’.
    - Sales = SUM([OrderQuantity])

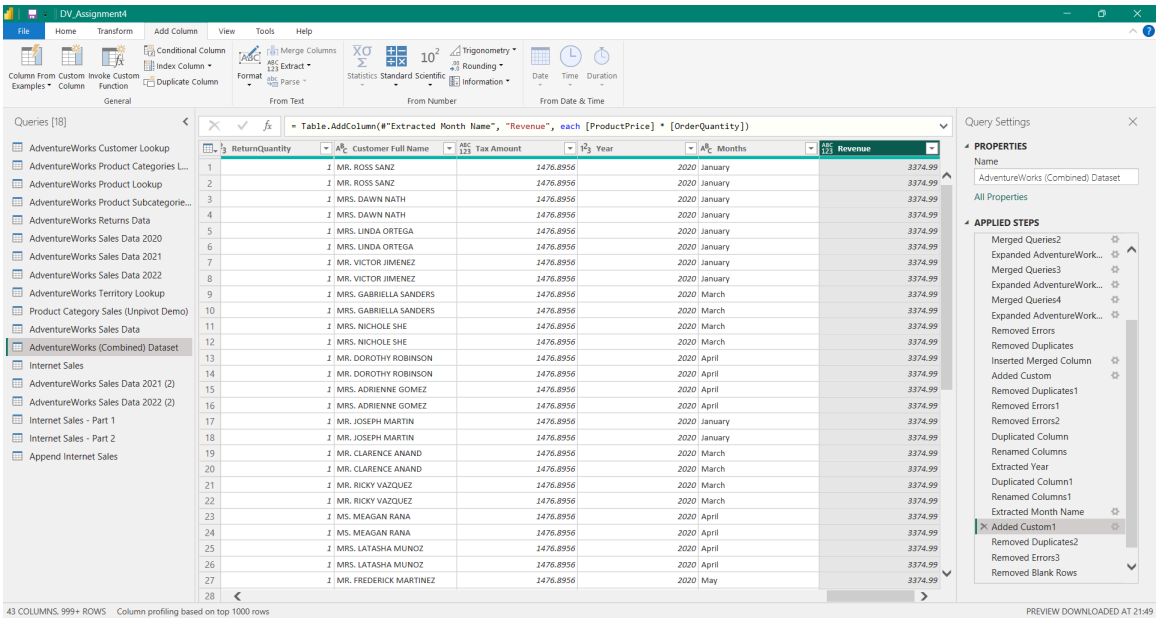




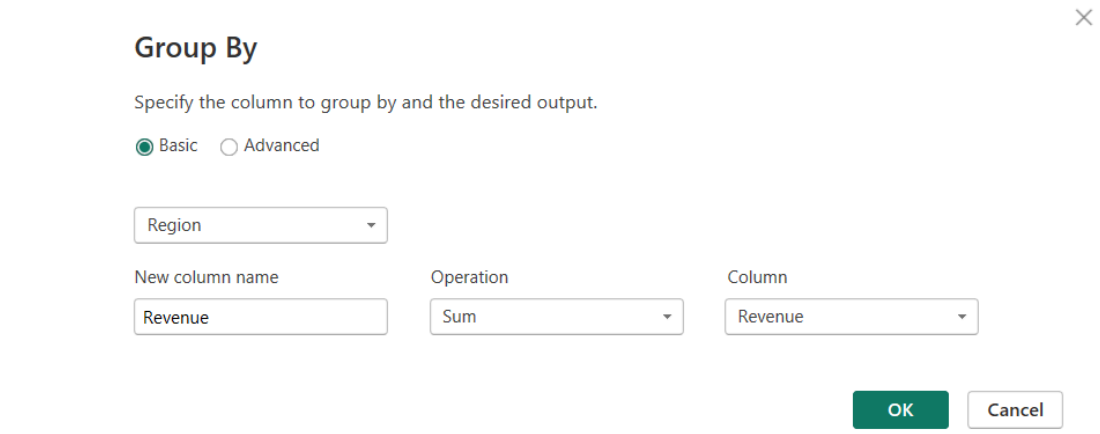
11. Determine which sales territory generated the highest revenue.

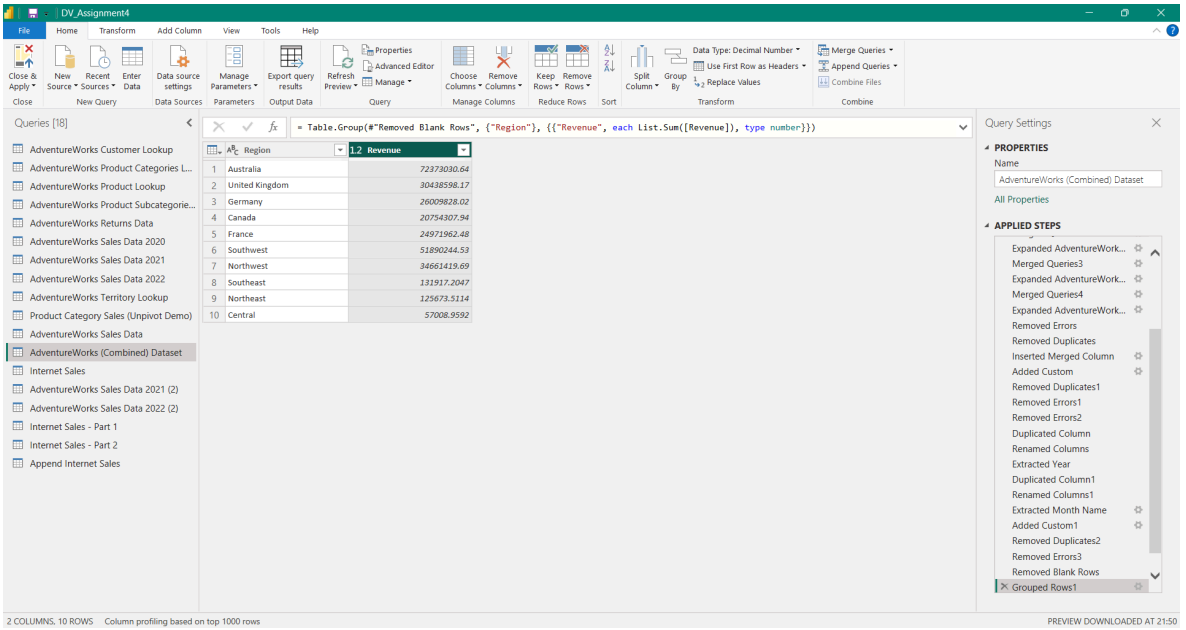
- Created a Custom Column “Revenue”
- Revenue = [ProductPrice] * [OrderQuantity]



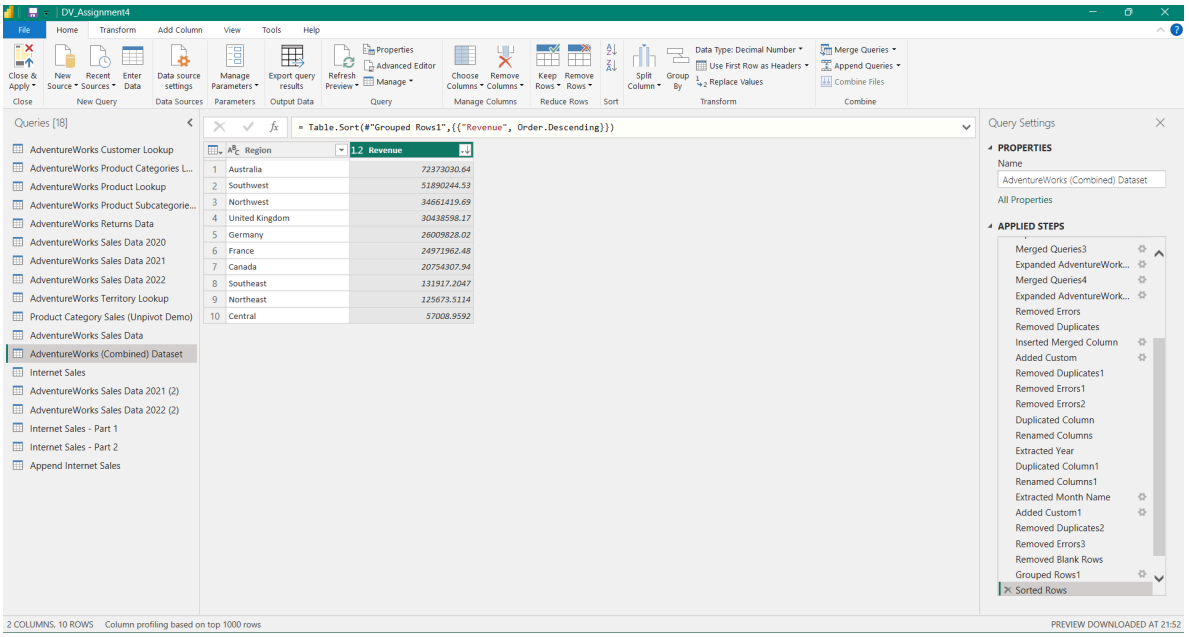


- Click on GroupBy which is on the Home / Transform tab. Select ‘Region’ for grouping ‘Revenue’.
- Revenue = SUM([Revenue])

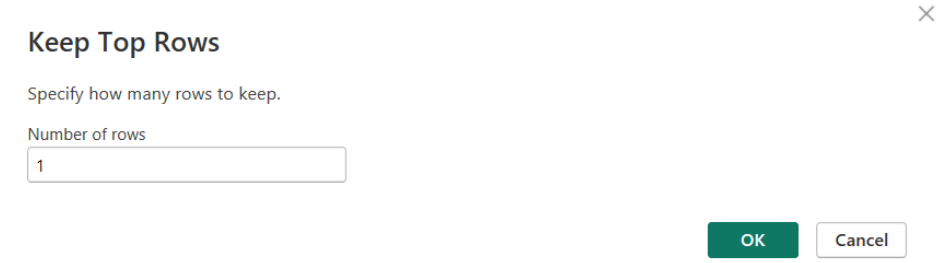


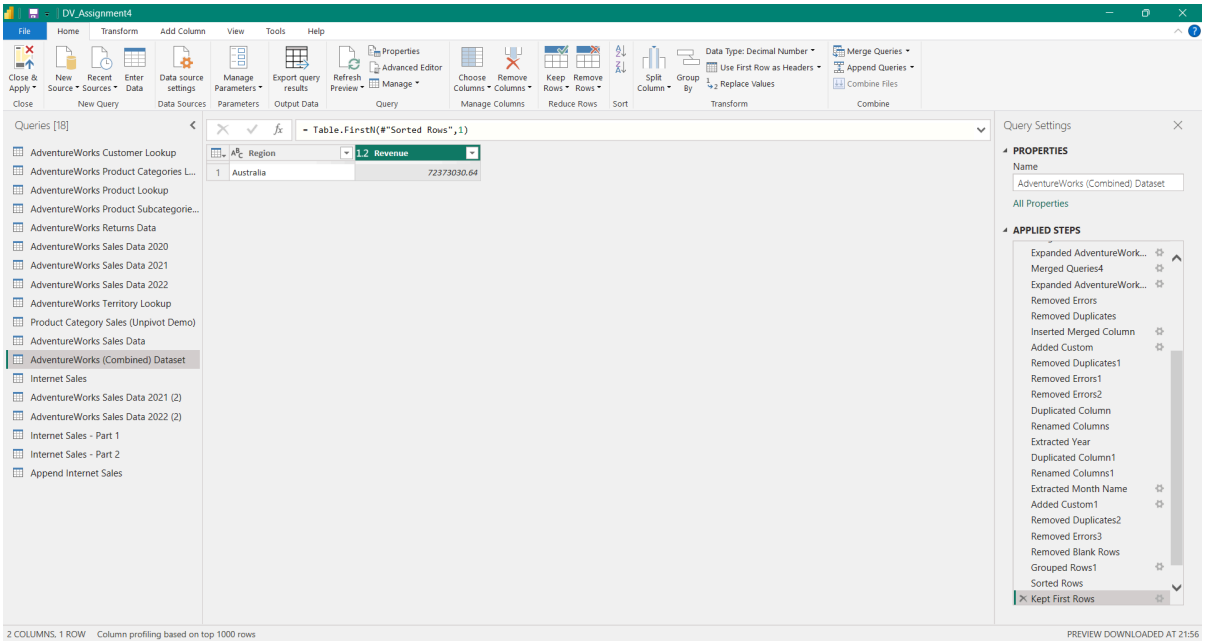


- Now, sort it in descending order by clicking on ‘Revenue’ and clicking on Sort (Z-A) from the Home tab.

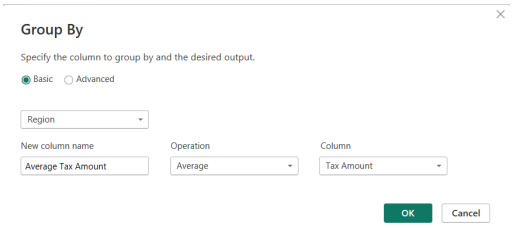


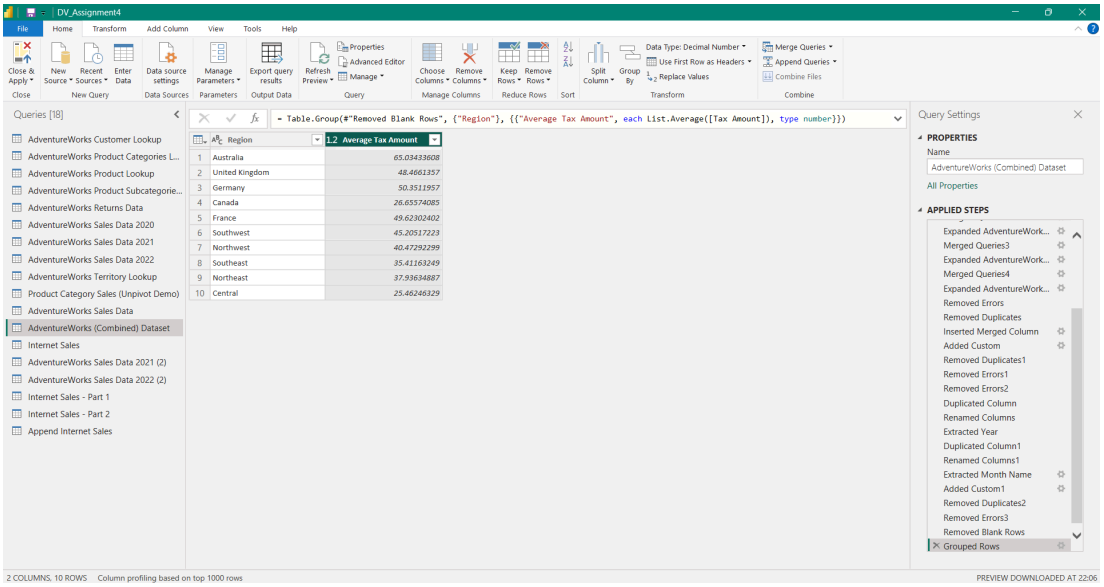
- Click on “Keep Rows” - Top 1 row





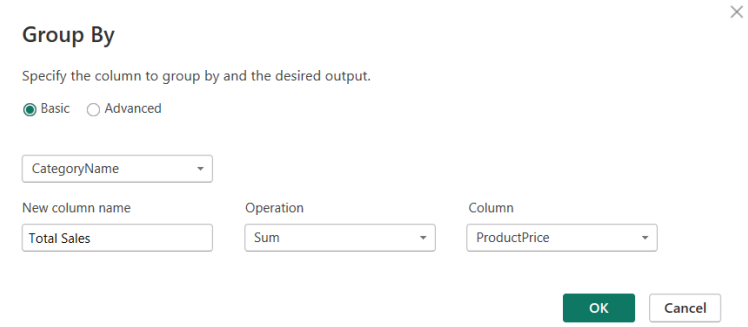
12. Find the average tax amount collected per territory.
    - Click on GroupBy which is on the Home / Transform tab. Select ‘Region’ for grouping ‘Average Tax Amount’.
    - Average Tax Amount = AVERAGE([Tax Amount])

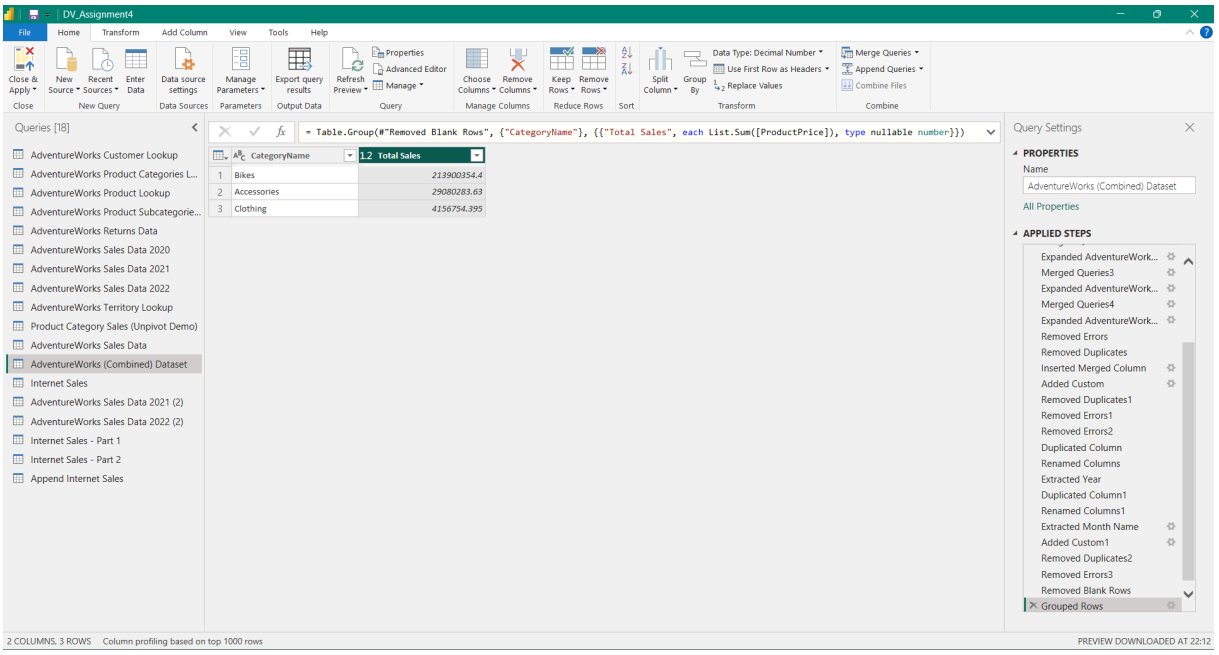




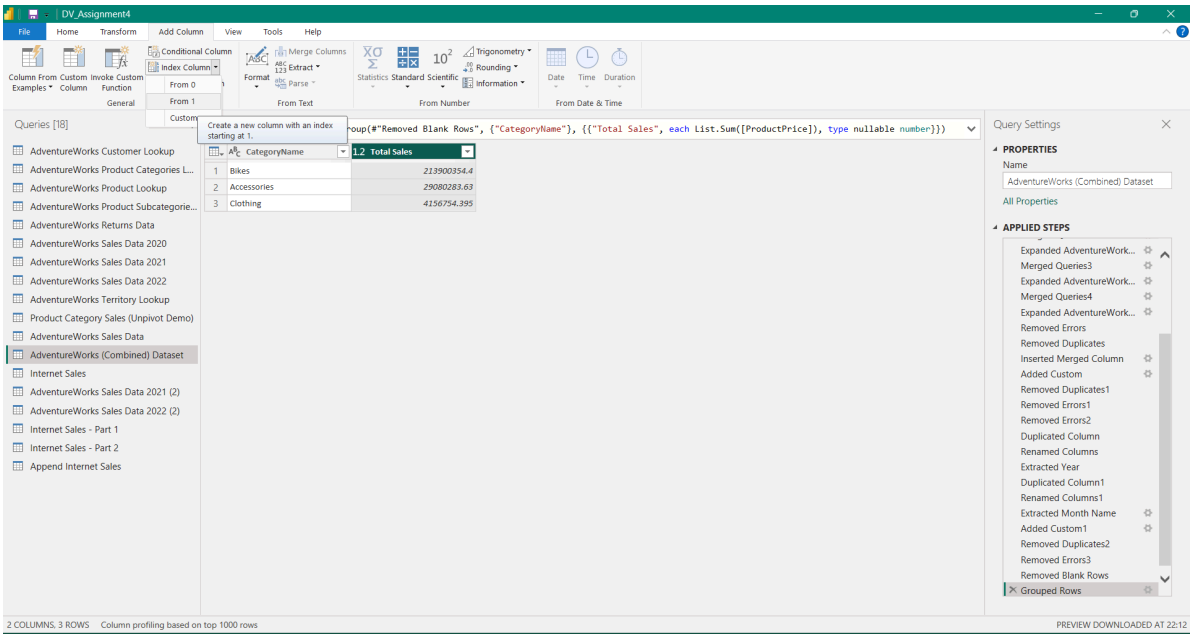
Advanced Tasks

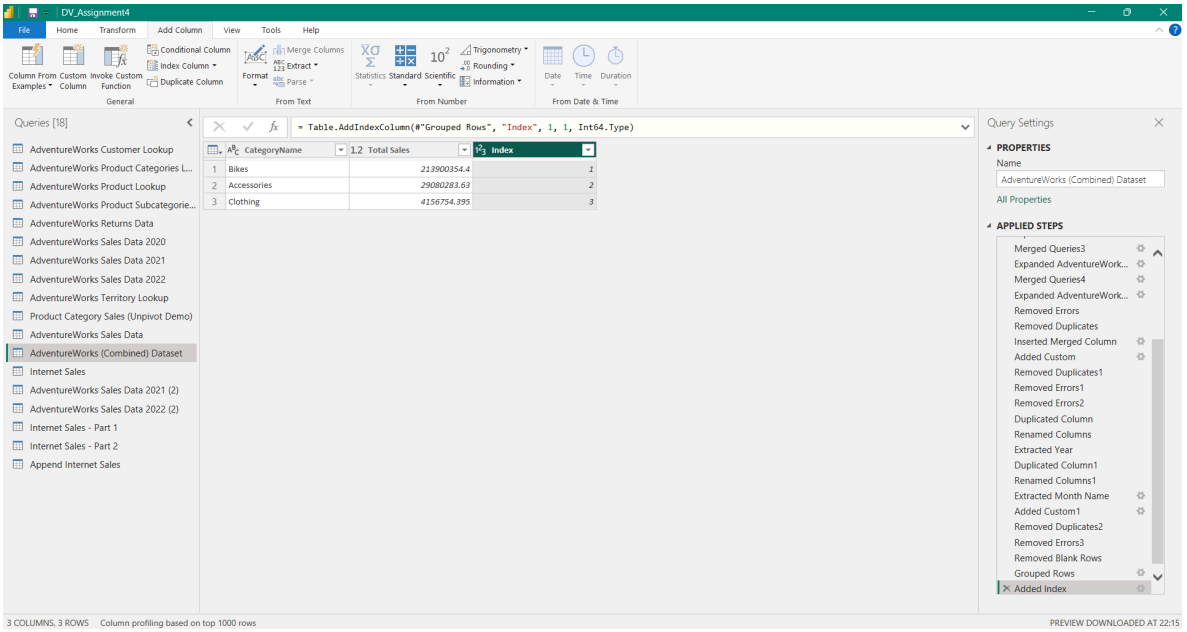
1. Rank product categories based on total sales.
    - Click on GroupBy which is on the Home / Transform tab. Select ‘CategoryName’ for grouping ‘Total Sales’.
    - Total Sales = SUM([ProductPrice])



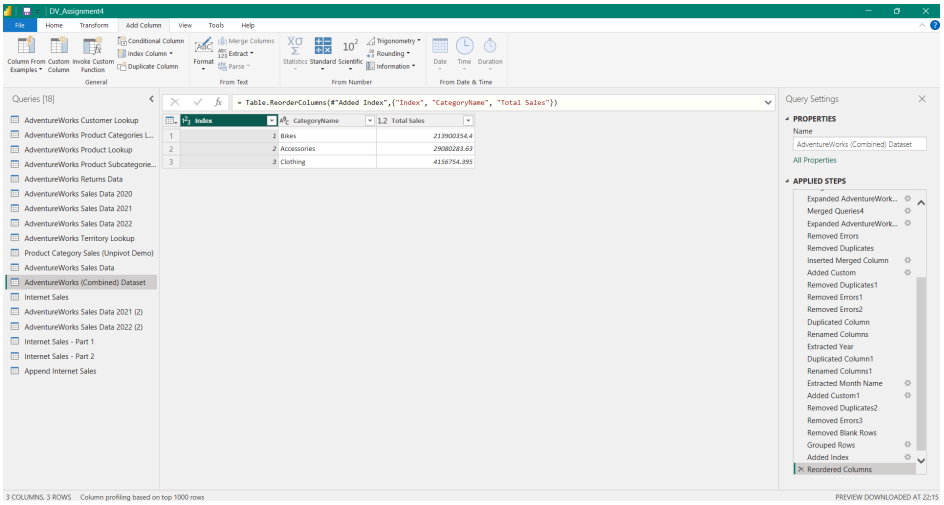


- Create an “Index” column by clicking on the “Total Sales” column and then clicking on ‘Index Column’ from the ‘Add Column’ tab.

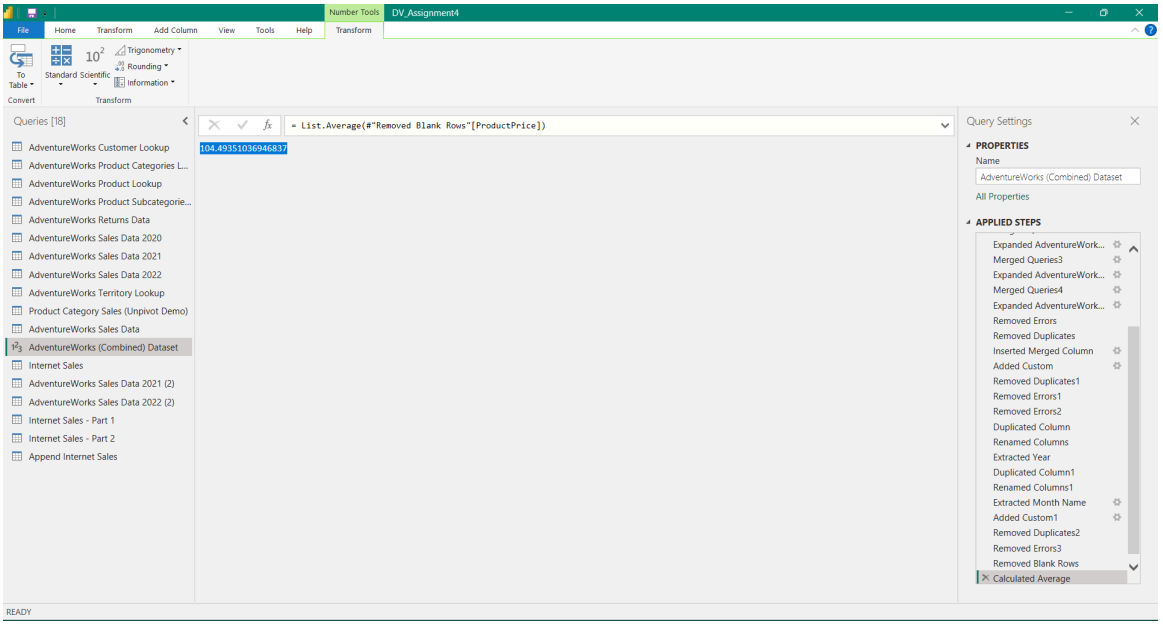




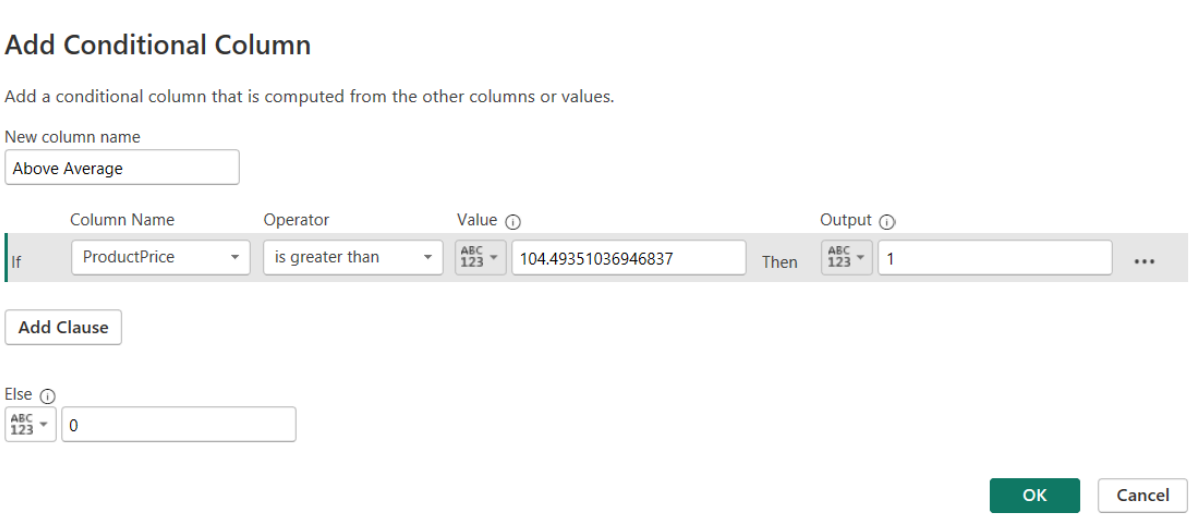
- Reorder the columns

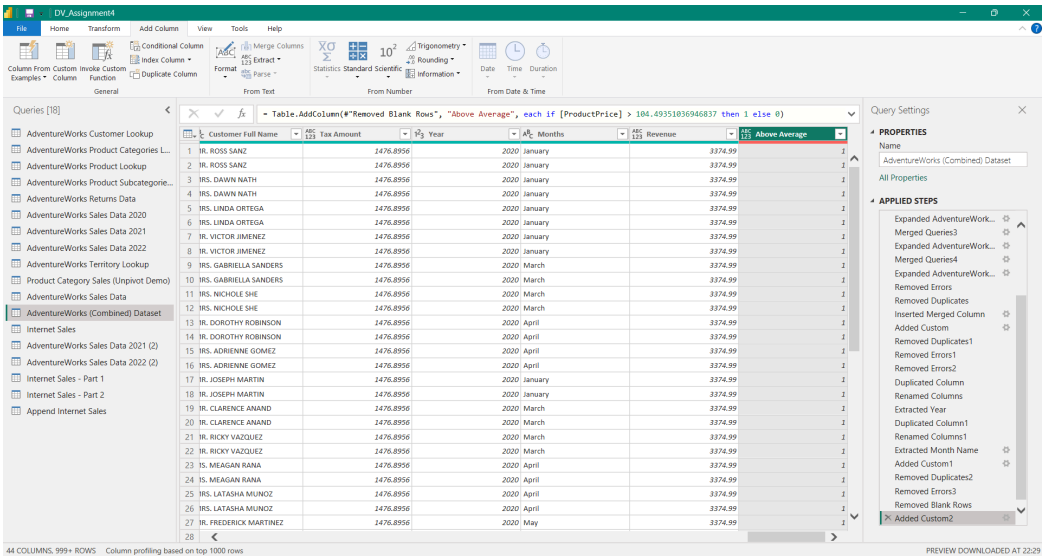


2. Identify products with above-average sales.
    - Find out "Average ProductPrice" by clicking on 'ProductPrice' column and then clicking on the Transform tab -> Statistics -> Average.

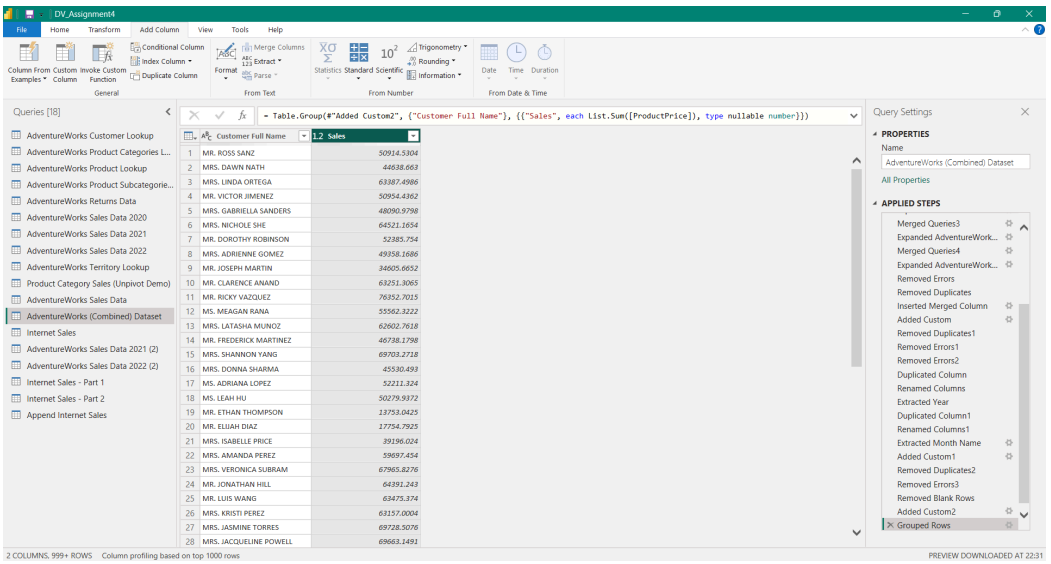


- Created conditional column - “Above Average”.
- Above Average = IF [ProductPrice] is greater than 104.49351036946873 THEN 1 ELSE 0

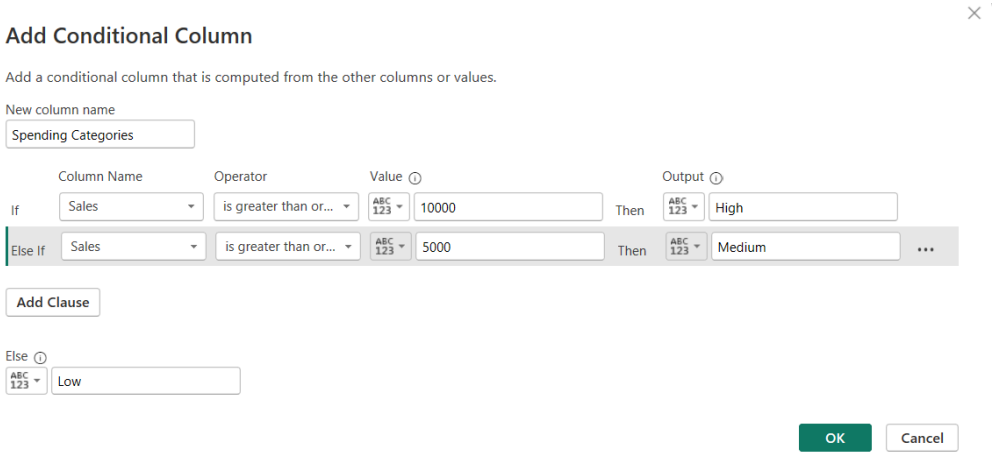


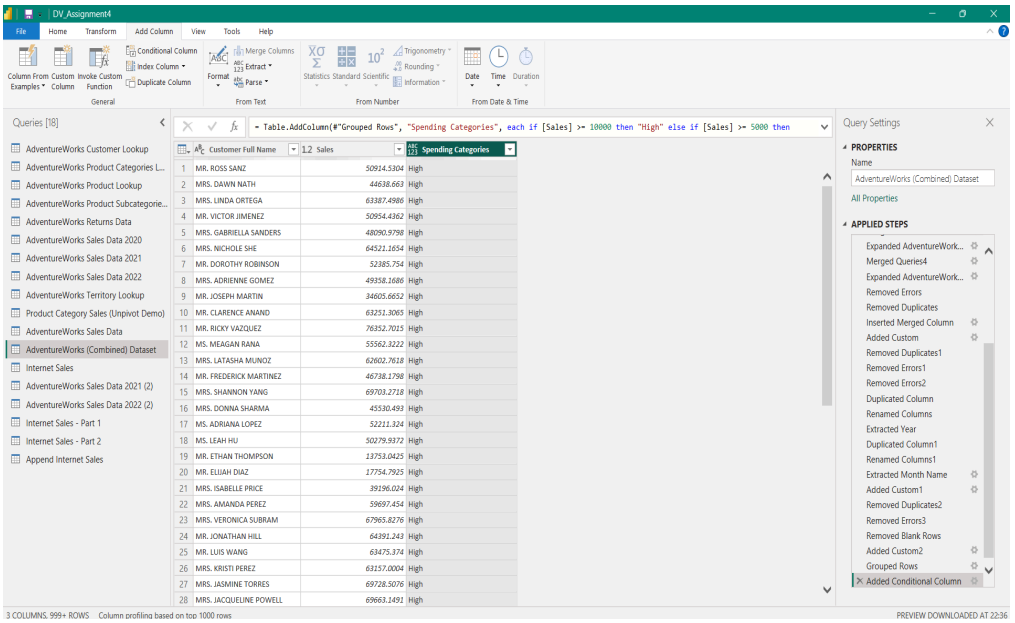


3. Group customers into High, Medium, and Low spending categories using Conditional Columns.
    - Click on GroupBy which is on the Home / Transform tab. Select ‘Customer Full Name’ for grouping ‘Sales’.
    - Sales = SUM([ProductPrice])

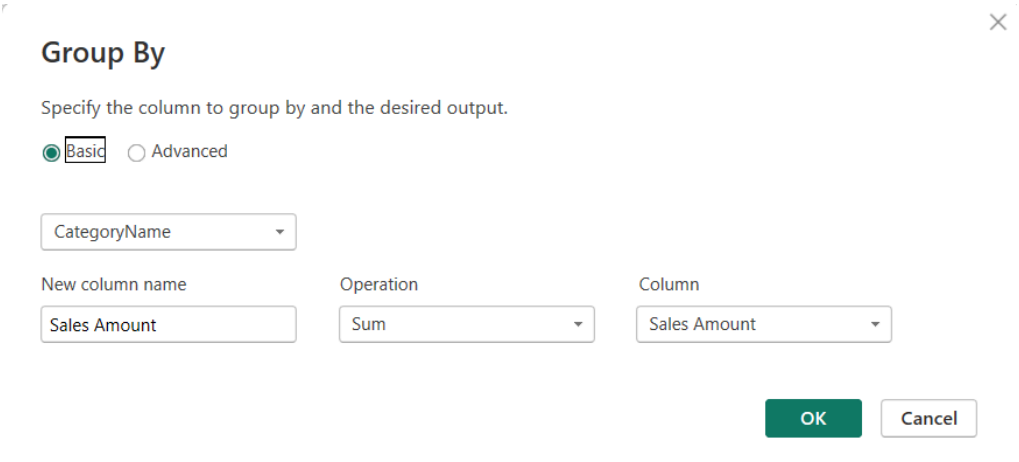


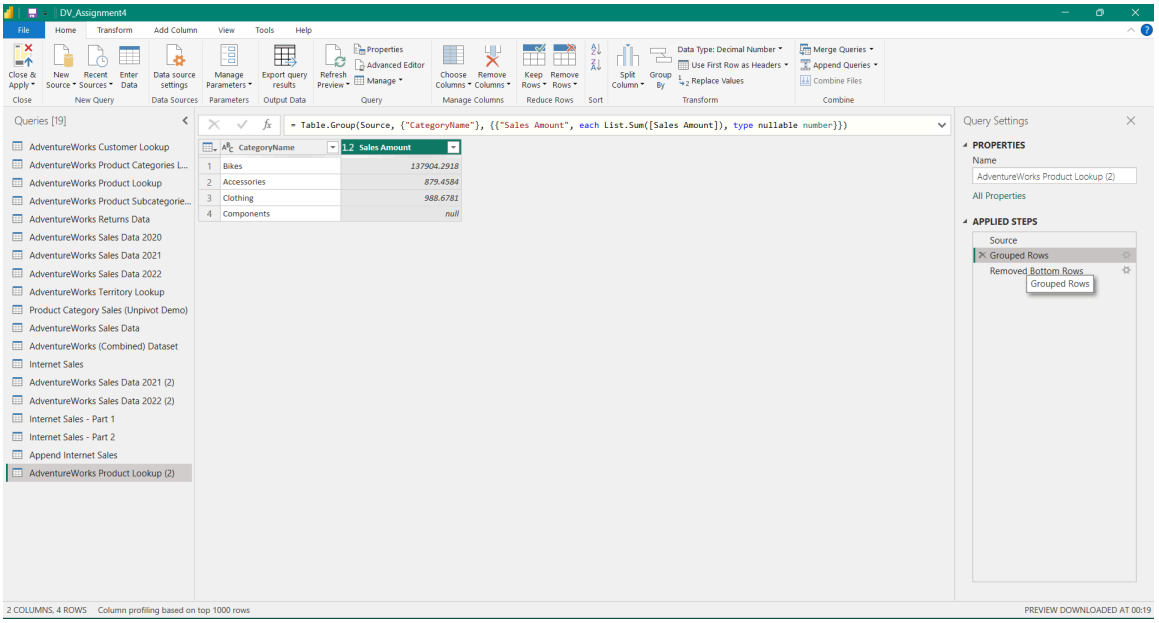
- Created conditional column - “Spending Categories”.
- Spending Categories = IF [Sales] is greater than or equal to 10000 THEN “High” ELSE IF [Sales] is greater than or equal to 5000 THEN “Medium” ELSE “Low”



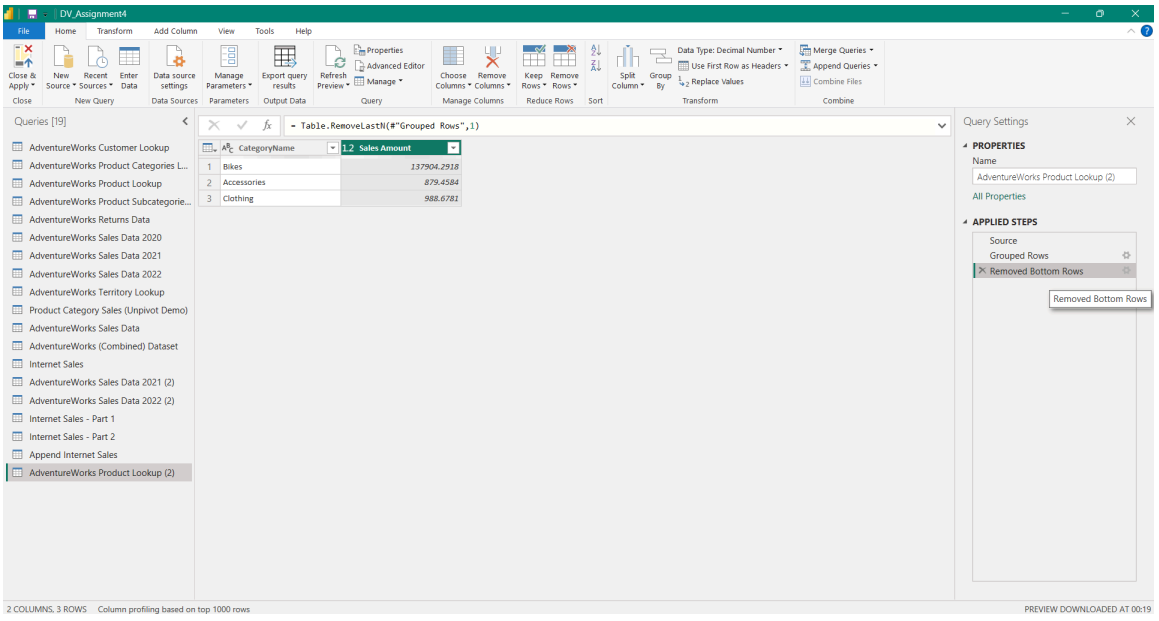


4. Calculate the percentage contribution of each product category to total sales.
    - Click on GroupBy which is on the Home / Transform tab. Select ‘CategoryName’ for grouping ‘Sales Amount’.
    - Sales Amount = SUM([Sales Amount])

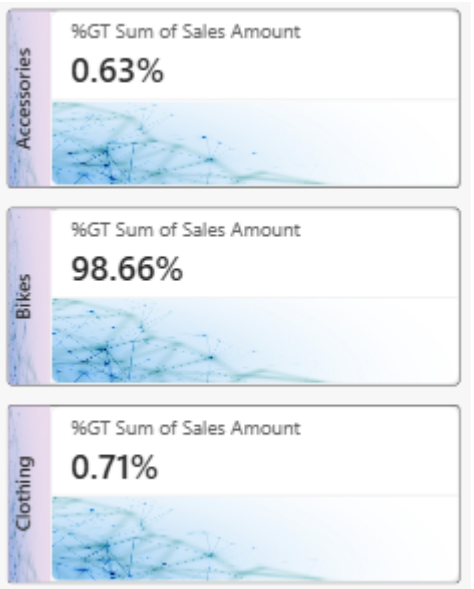




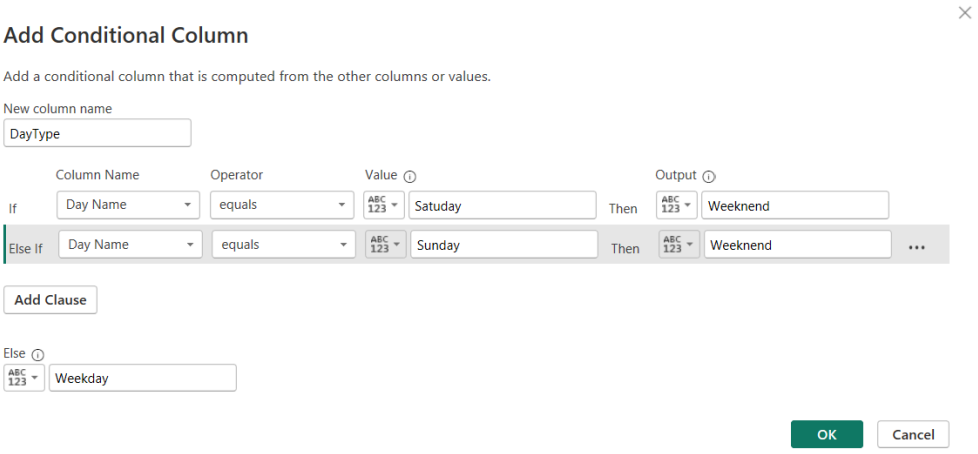
- Delete the last row containing the “null” value.



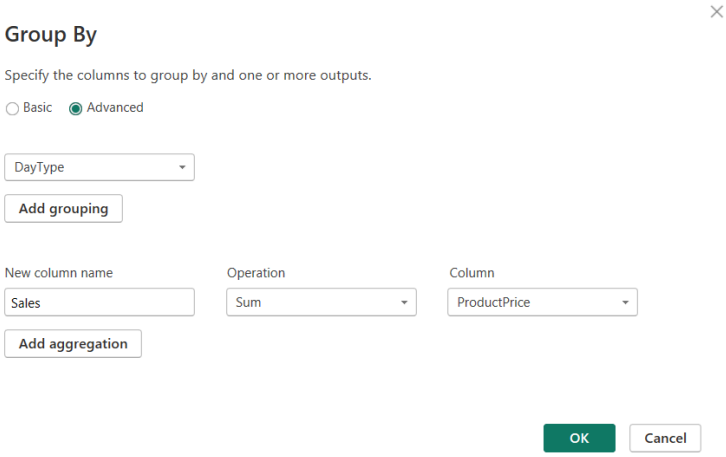
- Save and visualize using card visualization.

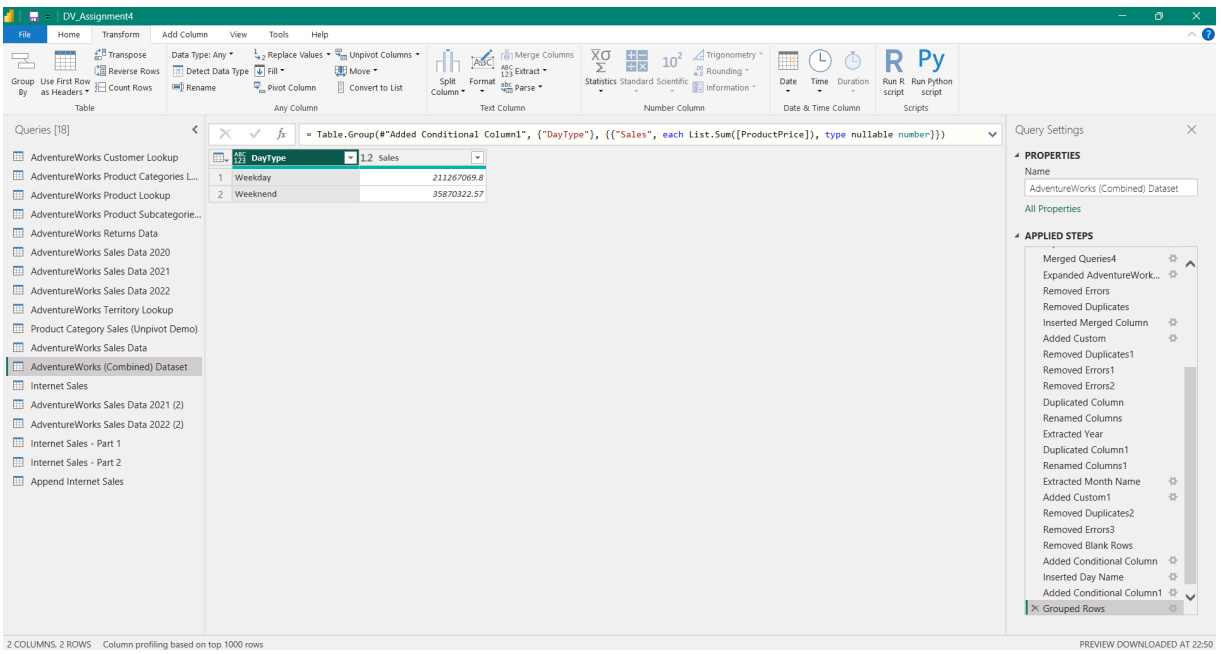


5. Compare weekend vs weekday sales by extracting the day of the week from the order date.
    - Created a conditional column - "DayType".
    - DayType = IF [Day Name] equals “Saturday” THEN “Weekend” ELSE IF [Day Name] equals “Saturday” THEN “Weekend” ELSE “Weekday”.

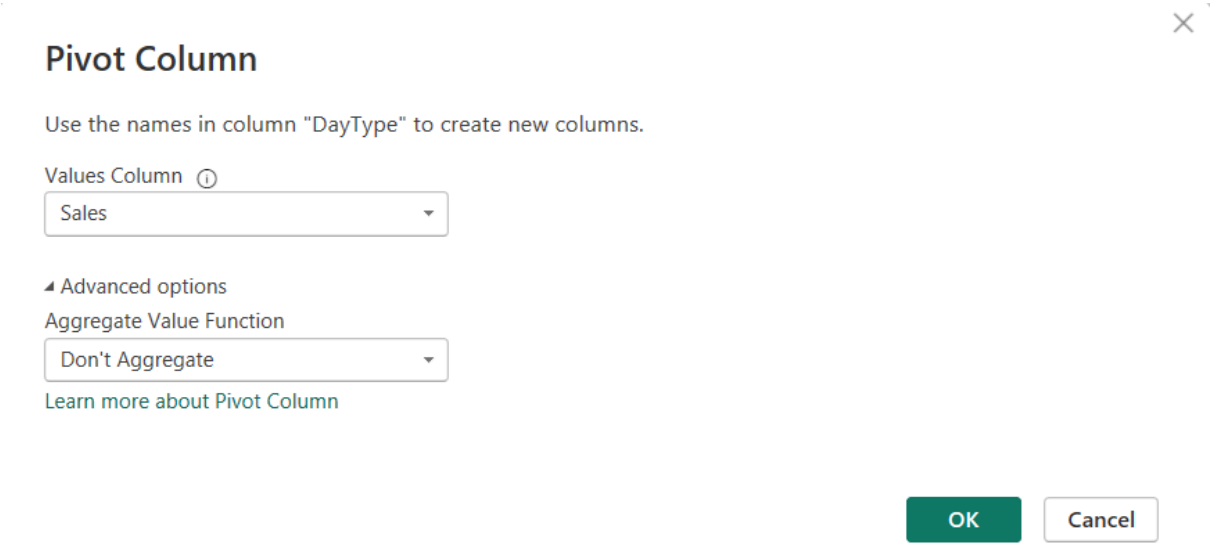


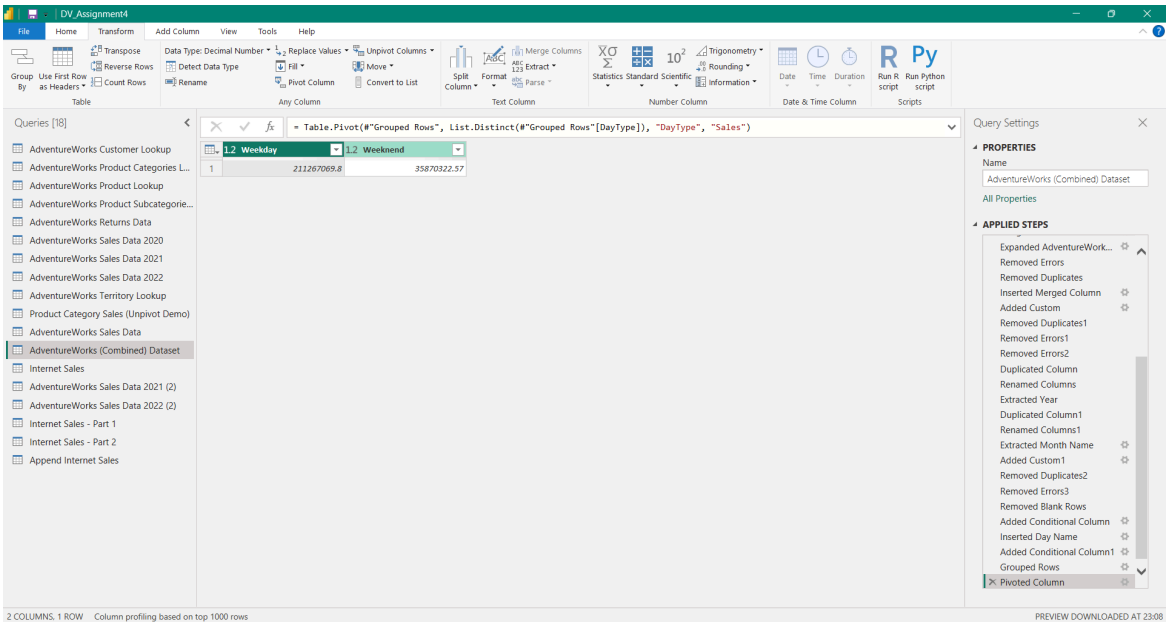
    - Click on GroupBy which is on the Home / Transform tab. Select ‘DayType’ for grouping ‘Sales Amount’.
    - Sales = SUM([ProductPrice])



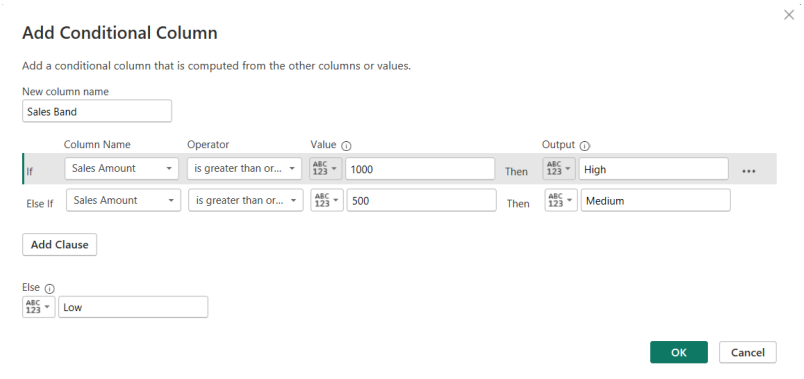


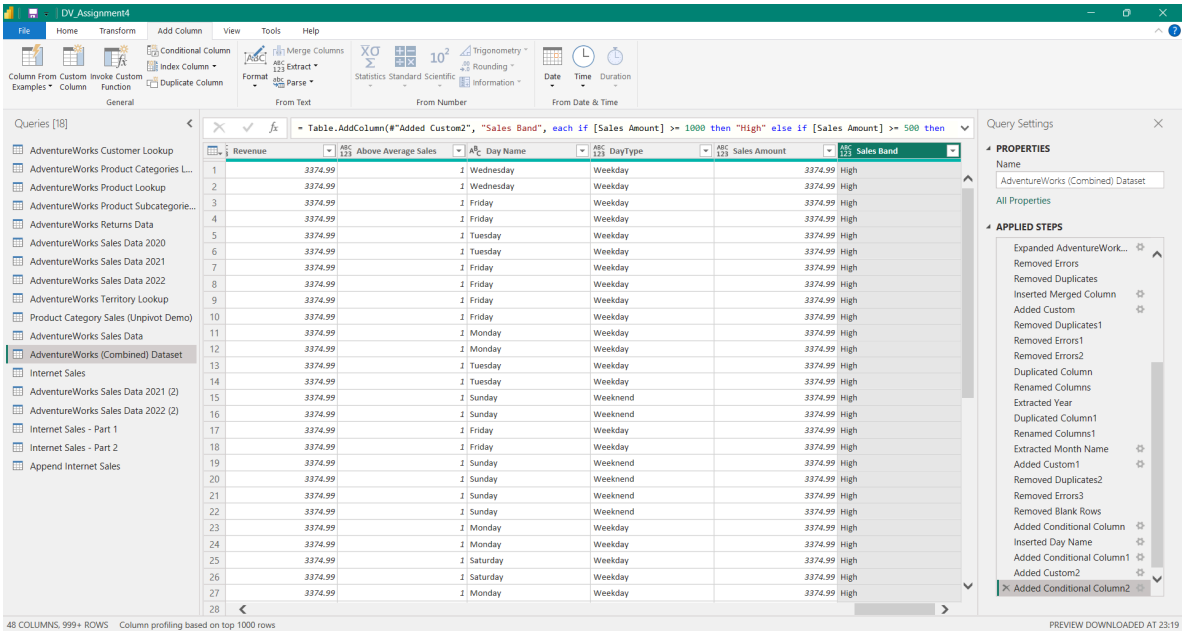
- Pivot “Sales” column using “DayType”.
- Choose “Don’t Aggregate”.





6. Create sales bands (Low, Medium, High) based on Sales Amount using Conditional Columns.
    - Created conditional column - “Spending Categories”.
    - Spending Categories = IF [Sales Amount] is greater than or equal to 1000 THEN “High” ELSE IF [Sales Amount] is greater than or equal to 500 THEN “Medium” ELSE “Low”





LINKS TO BLOG:

Platform	Link
Google Docs	https://docs.google.com/document/d/1nm1xQIAKoMqi2b4ggpeZISJ5UwNRcpgcyOHhbZ-H29Q/edit?usp=sharing
LinkedIn	https://www.linkedin.com/posts/mahita-suresh-thundiyil_dvassignment4-activity-7497352859349536768-_nZZ?utm_source=share&utm_medium=member_desktop&rcm=ACoAAFWEjDwBMzVflgHqsDSyNR2L6ydFzgdDLfg
GitHub	https://github.com/mst-2005/DataVisualization/tree/main/Assignment4